

MarketPeak_Ecommerce Project Report

Author: Oluwaseun Osunsola

Environment: Kali Linux (CLI-based)

Project link: <https://>

Project Introduction

This capstone project focuses on building and deploying an **e-commerce platform** named **MarketPeak**. The platform will provide core online shopping functionalities such as product listings, a shopping cart, and user authentication. The project demonstrates end-to-end web development and deployment practices, starting from selecting and customizing a suitable HTML/CSS template to hosting the website on an AWS EC2 instance.

The entire development process is tracked with **Git** for version control, ensuring proper management of changes, collaboration, and rollback capabilities. The implementation takes place in a **Linux environment**, reinforcing industry-standard deployment workflows. Finally, the live platform is made accessible via an **Apache web server** running on a cloud-based infrastructure.

Project Background

MarketPeak is envisioned as a scalable online marketplace that can grow with the needs of vendors and customers. The project's foundation lies in leveraging **ready-made HTML/CSS templates** from sources such as Tooplate, which allows rapid prototyping and customization without building the interface from scratch.

This project bridges three crucial aspects of modern web development:

1. **Version Control (Git & GitHub)** – Ensuring the codebase is organized, backed up, and collaboration-ready.
2. **Linux Server Administration** – Managing file systems, permissions, and service configuration for reliable deployment.
3. **Cloud Hosting (AWS EC2)** – Utilizing elastic, on-demand infrastructure to host and scale the application.

By integrating these components, MarketPeak serves as a comprehensive demonstration of **full-stack deployment** in a real-world setting.

Project Objectives

The objectives of this project are to:

1. **Select and Customize a Web Template** Adapt a Tooplate template to match the MarketPeak brand identity.
2. **Implement Git-based Version Control** Track all changes, manage branches, and push updates to a remote GitHub repository.

3. **Set Up a Linux Development and Hosting Environment** Use Ubuntu Server for local development and AWS EC2 for production hosting.
 4. **Deploy the E-Commerce Platform to AWS EC2** Configure Apache to serve the website publicly over HTTP/HTTPS.
 5. **Demonstrate a Live Update Workflow** Make code changes in a development branch, merge via GitHub pull requests, and push updates to the live server.
 6. **Document the Process in Detail** Provide a step-by-step report, supported with 58 screenshots, to ensure the workflow can be replicated by others.
-

Step 1: Create Project Folder and Initialize Git

Open your terminal, navigate to the desired parent directory, and create the project folder. Git will initialize the empty repository.

- **Command:**

```
mkdir MarketPeak_Ecommerce
cd MarketPeak_Ecommerce
git init
```

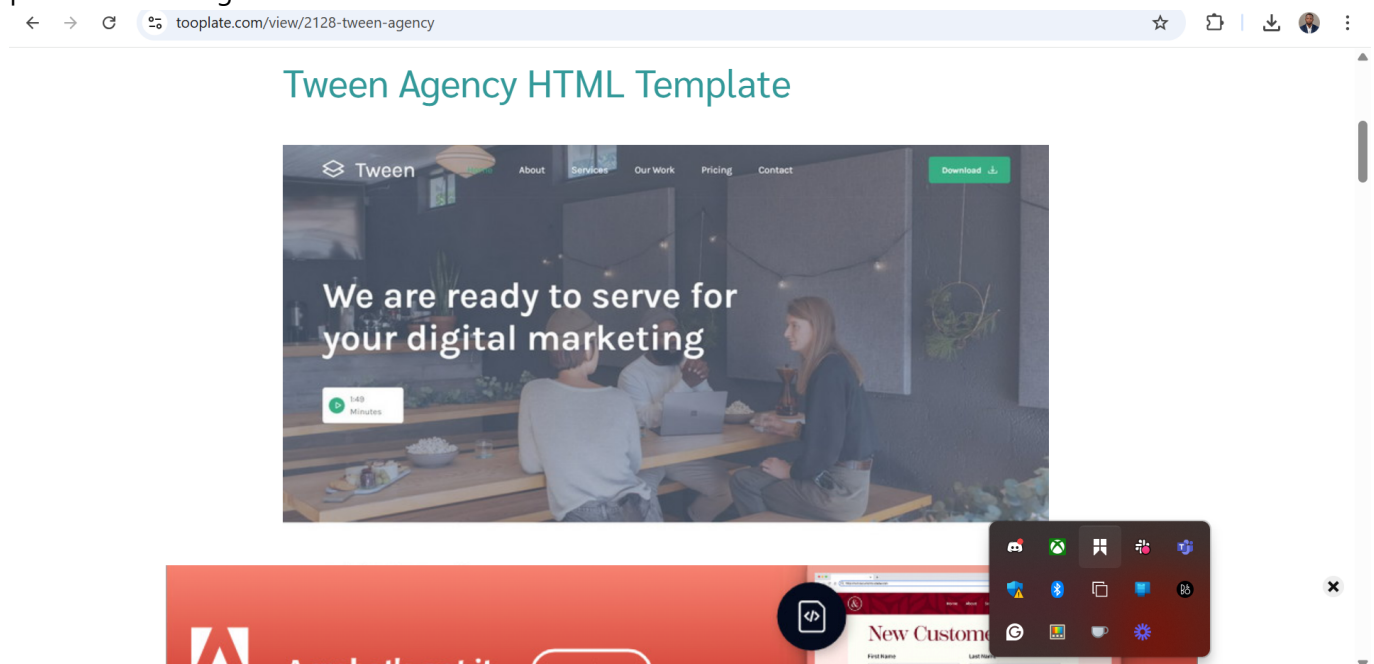
```
HP@DESKTOP-19W74R1 MINGW64 ~/Documents/workspace
$ mkdir MarketPeak_Ecommerce

HP@DESKTOP-19W74R1 MINGW64 ~/Documents/workspace
$ cd MarketPeak_Ecommerce/

HP@DESKTOP-19W74R1 MINGW64 ~/Documents/workspace/MarketPeak_Ecommerce
$ git init
```

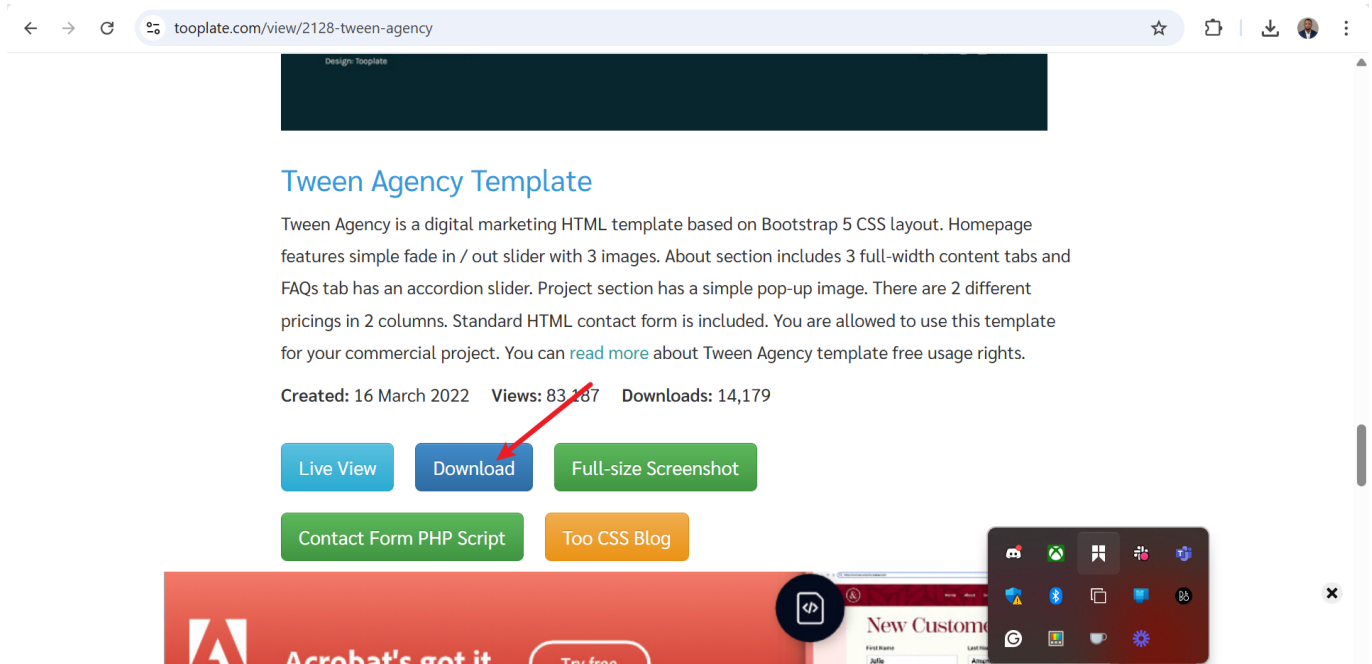
Step 2: Pick a Tooplate Website Template

Visit [Tooplate.com](https://tooplate.com) and select a suitable e-commerce template. This saves development time and provides a professional design base.



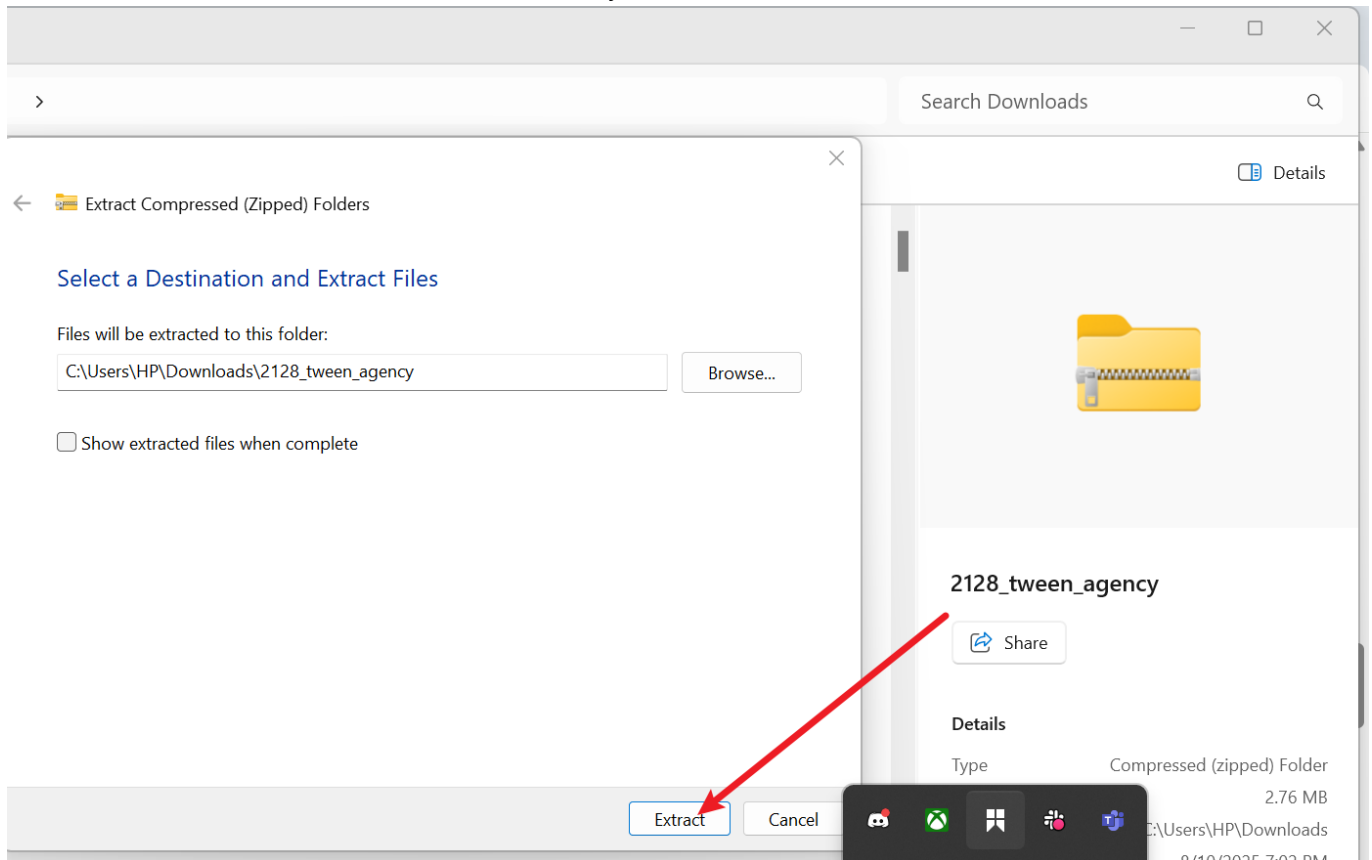
Step 3: Download the Website Template

Click the download button in order to download a copy of the selected template.



Step 4: Extract the Downloaded Template

Locate the ZIP file of the downloaded website in your chosen download folder and extract it.



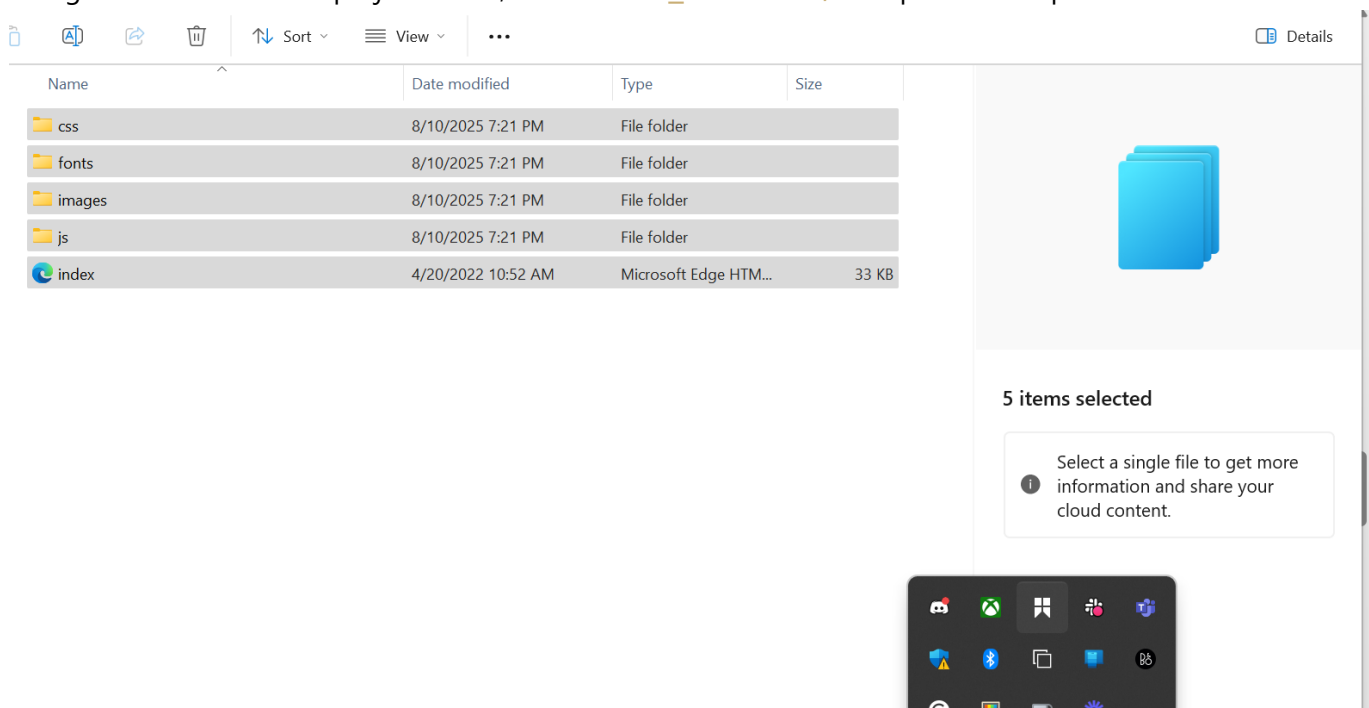
Step 5: Copy/Cut Template Files and Assets

Open the extracted folder, select all the files in there, and copy or cut them.

Name	Date modified	Type	Size
▼ Today			
css	8/10/2025 7:21 PM	File folder	
fonts	8/10/2025 7:21 PM	File folder	
images	8/10/2025 7:21 PM	File folder	
js	8/10/2025 7:21 PM	File folder	
▼ A long time ago			
index	4/20/2022 10:52 AM	Microsoft Edge HTM...	33 KB

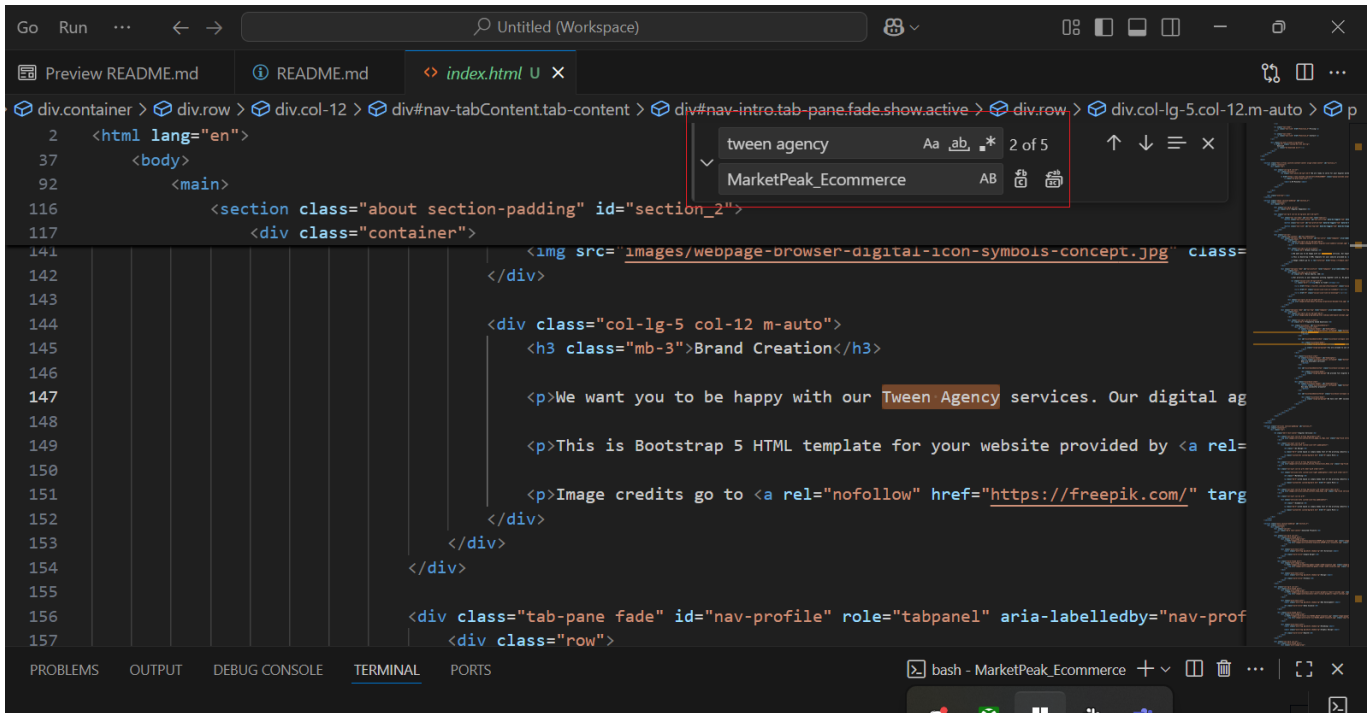
Step 6: Paste Files into the Project Folder

Navigate to the initialized project folder, **MarketPeak_Ecommerce/** and paste the copied files into it.



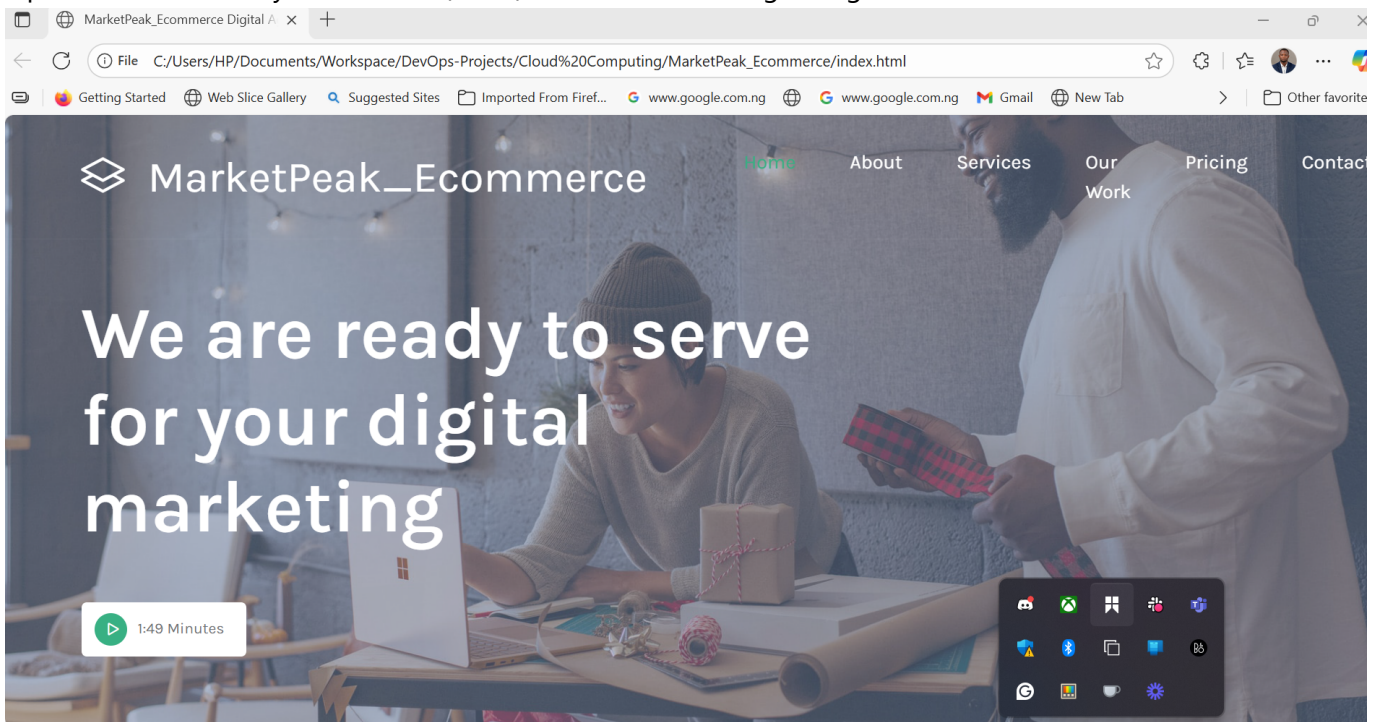
Step 7: Customize Website Name

Open `index.html` in a code editor (VS Code, Sublime Text, etc.) and change the `<title>` tag and any visible branding to "MarketPeak_Ecommerce".



Step 8 — Preview site locally after customization

Open `index.html` in your browser (file://) to confirm branding changes.



Step 9 — Stage the website files for Git

Initialize the project folder and add all project files to Git staging area.

- **Command:**

```
git add .
```

`git status` shows files in "Changes to be committed".

```
HP@DESKTOP-19M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce
$ git init
Initialized empty Git repository in C:/Users/HP/Documents/Workspace/MarketPeak_Ecommerce/.git/

HP@DESKTOP-19M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (main)
$ git add .
warning: in the working copy of 'css/bootstrap-icons.css', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'css/bootstrap.min.css', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'css/magnific-popup.css', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/bootstrap.min.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/click-scroll.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/custom.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/jquery.backstretch.min.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/jquery.magnific-popup.min.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/jquery.min.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/magnific-popup-options.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'js/popper.js', LF will be replaced by CRLF the next time Git touches it
```

Step 10 — Set git global configuration (name & email)

Configure Git identity so commits are properly attributed.

- **Commands:**

```
git config --global user.name "Ogunsola Oluwaseun"
git config --global user.email "oluwaseun.beicks@gmail.com"
```

```
HP@DESKTOP-19M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (main)
$ git config --global user.name "Oluwaseun Ogunsola"

HP@DESKTOP-19M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (main)
$ git config --global user.email "oluwaseun.beicks@gmail.com"

HP@DESKTOP-19M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (main)
$
```

Step 11 — Commit changes with descriptive message

Create the first (initial) commit with a helpful message.

- **Command:**

```
git commit -m "Initial commit of basic ecommerce structure"
```

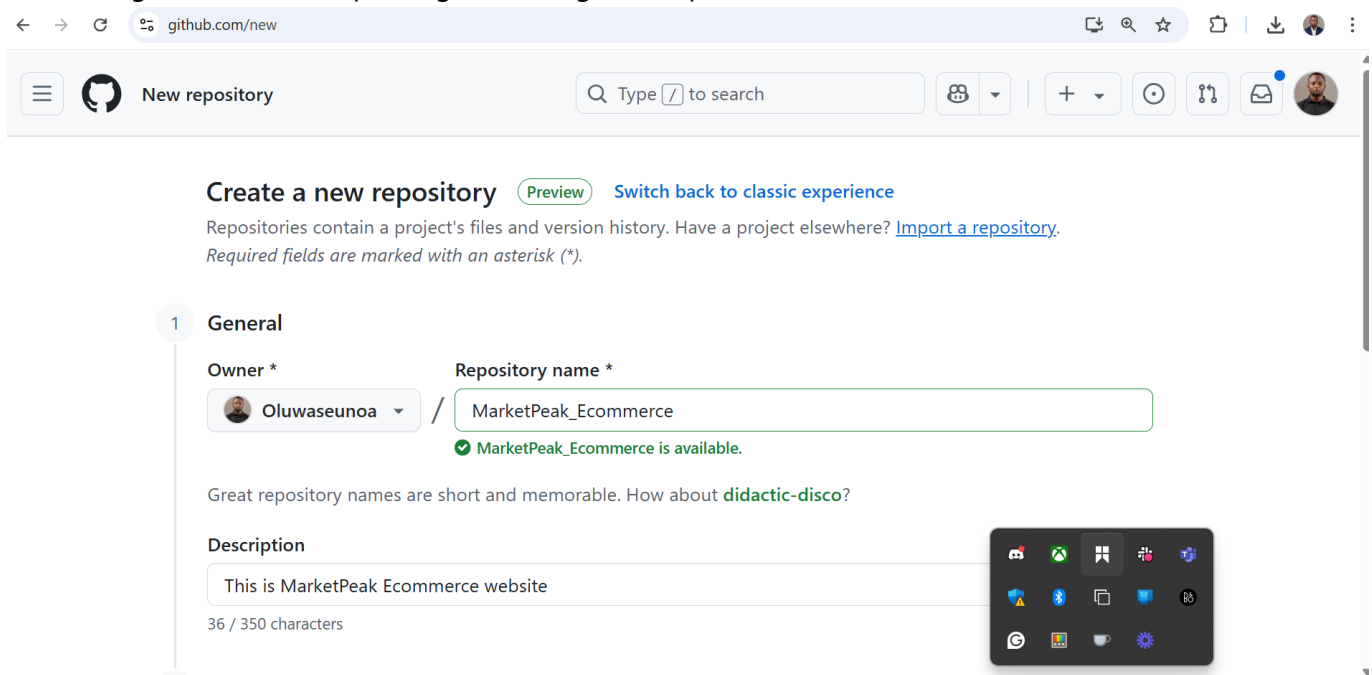
```

$ git commit -m "Initial commit of with basic ecommerce site structure"
[main (root-commit) bbaef2b] Initial commit of with basic ecommerce site structure
30 files changed, 5740 insertions(+)
create mode 100644 css/bootstrap-icons.css
create mode 100644 css/bootstrap.min.css
create mode 100644 css/magnific-popup.css
create mode 100644 css/tooplate-tween-agency.css
create mode 100644 fonts/bootstrap-icons.woff
create mode 100644 fonts/bootstrap-icons.woff2
create mode 100644 images/dream-about-working-corporation-became-true.jpg
create mode 100644 images/maria-stewart-p4tj0g_aMM-unsplash.jpg
create mode 100644 images/php-programming-html-coding-cyberspace-concept.jpg
create mode 100644 images/services/undraw_online_page_re_lhgx.svg
create mode 100644 images/services/undraw_content_team_3epn.svg
create mode 100644 images/services/undraw_online_transactions_02ka.svg
create mode 100644 images/slide/microsoft-edge-FAaz8lkinzs-unsplash.jpg
create mode 100644 images/slide/surface-1x5jnh1p3Y-unsplash.jpg
create mode 100644 images/slide/surface-71_s6R0JpGc-unsplash.jpg
create mode 100644 images/webpage-browser-digital-icon-symbols-concept.jpg
create mode 100644 images/work/anthony-espinosa-pVQSM-p_0_c-unsplash.jpg
create mode 100644 images/work/business-chart-visual-graphics-report-concept.jpg
create mode 100644 images/work/mos-sukjaroenkraisri-j28Am0Dyhig-unsplash.jpg
create mode 100644 images/work/samantha-gades-lf2pD--wTUA-unsplash.jpg
create mode 100644 images/work/tyler-nix-HmVQh_EQJHY-unsplash.jpg
create mode 100644 index.html
create mode 100644 js/bootstrap.min.js
create mode 100644 js/click-scroll.js
create mode 100644 js/custom.js
create mode 100644 js/jquery.backstretch.min.js
create mode 100644 js/jquery.magnific-popup.min.js
create mode 100644 js/jquery.min.js
create mode 100644 js/magnific-popup-options.js
create mode 100644 js/popper.js

```

Step 12 — Create GitHub remote repository

In GitHub UI create a new repo named **MarketPeak_Ecommerce** (leave it empty — no README). This way we avoid merge conflicts when pushing an existing local repo.



← → ↻ github.com/new

☰ GitHub New repository 🔍 Type / to search

Create a new repository [Preview](#) [Switch back to classic experience](#)

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *  Oluwaseunola

Repository name *

✔ MarketPeak_Ecommerce is available.

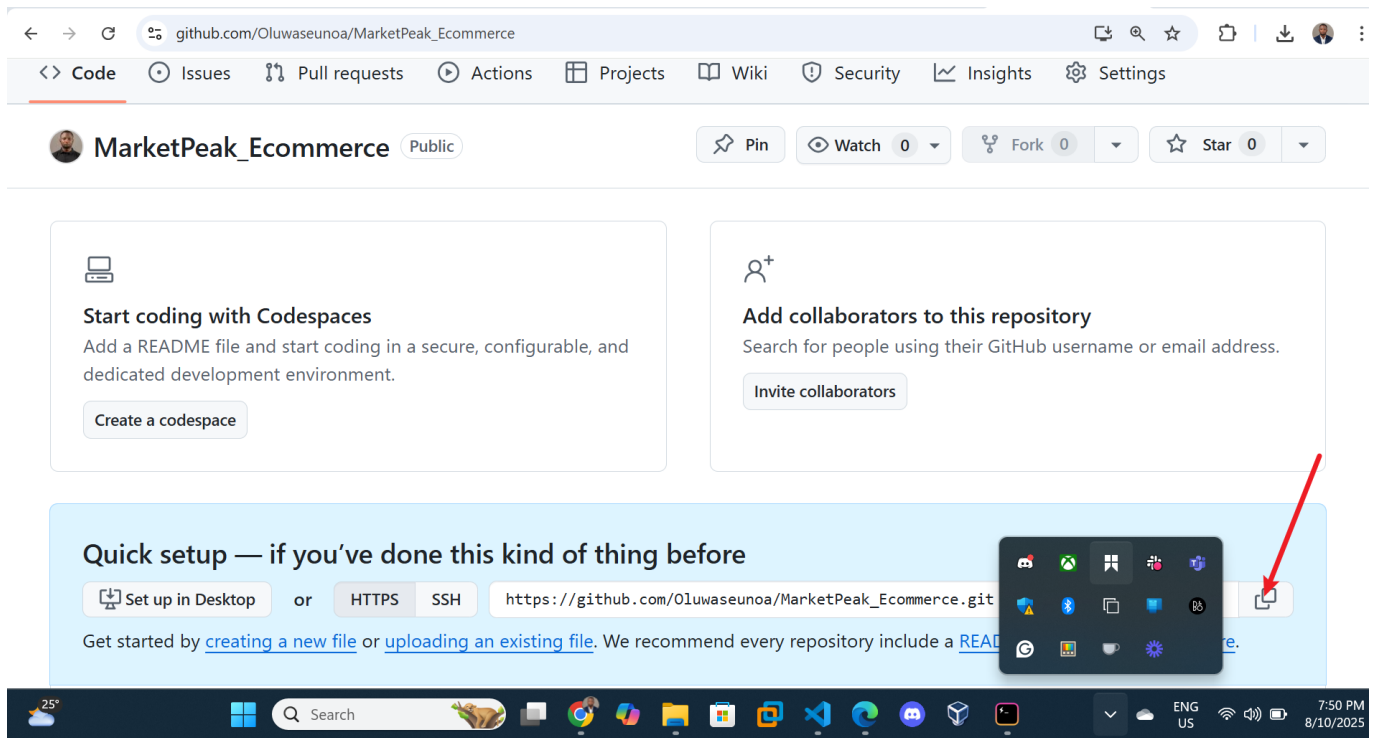
Great repository names are short and memorable. How about [didactic-disco](#)?

Description

36 / 350 characters

Step 13 — Copy repository link (HTTPS recommended)

In the **Quick setup** section, copy the HTTPS clone URL. My own looks like: `https://github.com/your-username/MarketPeak_Ecommerce.git`

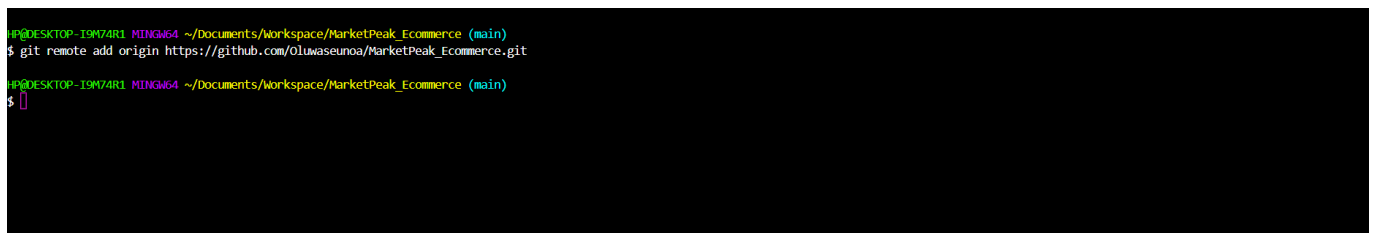


Step 14 — Add remote origin locally

In the local project repository, add the GitHub link copied to your project so that you can make the repository house your project on GitHub.

- **Command:**

```
git remote add origin https://github.com/your-username/MarketPeak_Ecommerce.git
```



Step 15 — Push main branch to GitHub

- **Commands:**

```
git push -u origin main
```

You will see something like

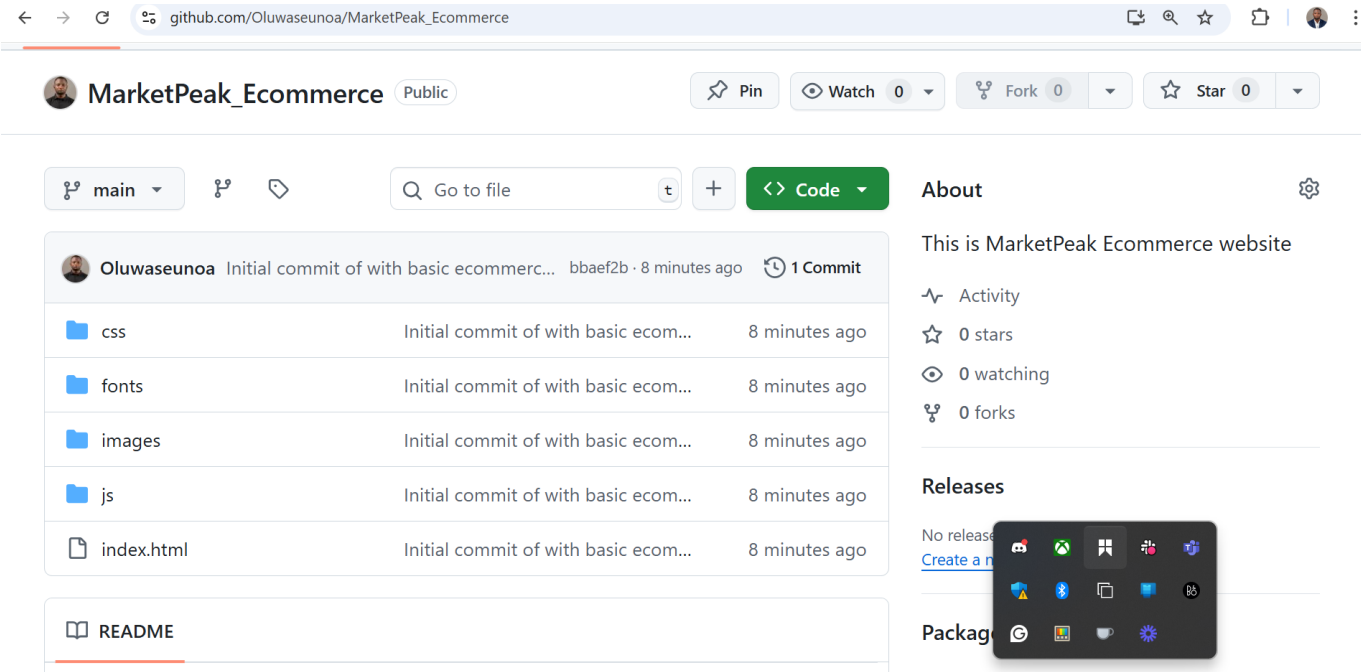
```
Progress bars and `To https://github.com/.../MarketPeak_Ecommerce.git` then  
`branch 'main' set up to track 'origin/main`.
```

```
HP@DESKTOP-19M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (main)
$ git push -u origin main
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 8 threads
Compressing objects: 100% (39/39), done.
Writing objects: 100% (39/39), 2.77 MiB | 334.00 KiB/s, done.
Total 39 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Oluwaseunola/MarketPeak_Ecommerce.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.

HP@DESKTOP-19M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (main)
$
```

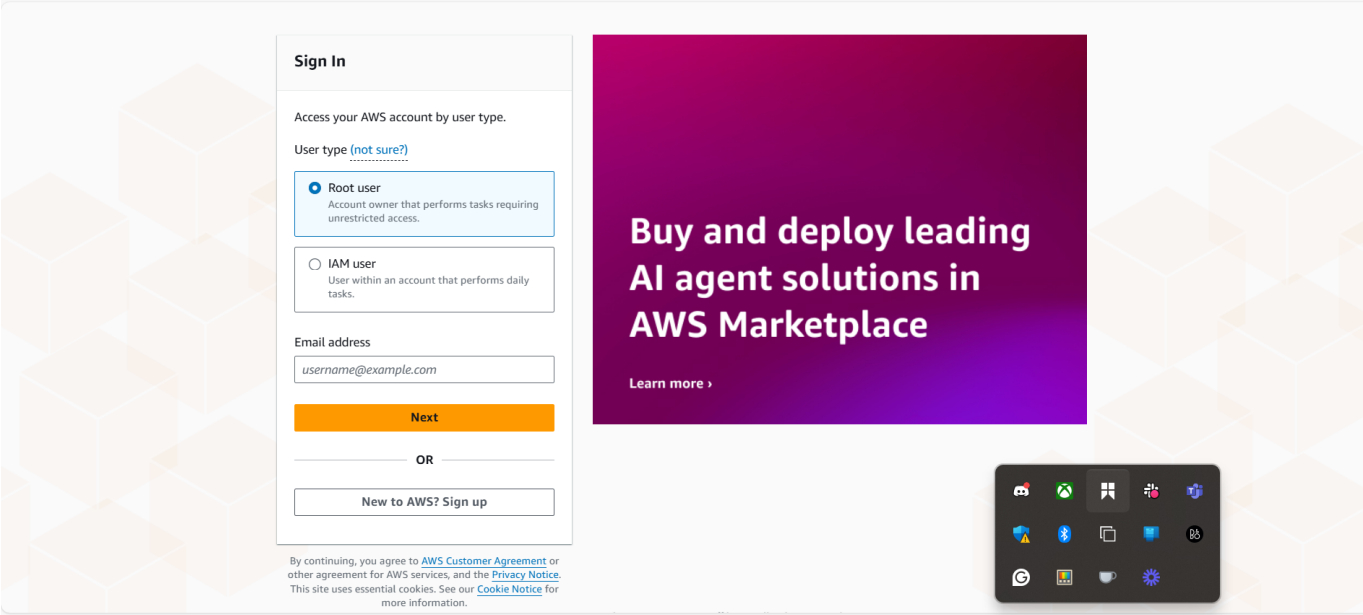
Step 16 — Verify files on GitHub

Open your GitHub repo in a browser to confirm files and commit history are present.



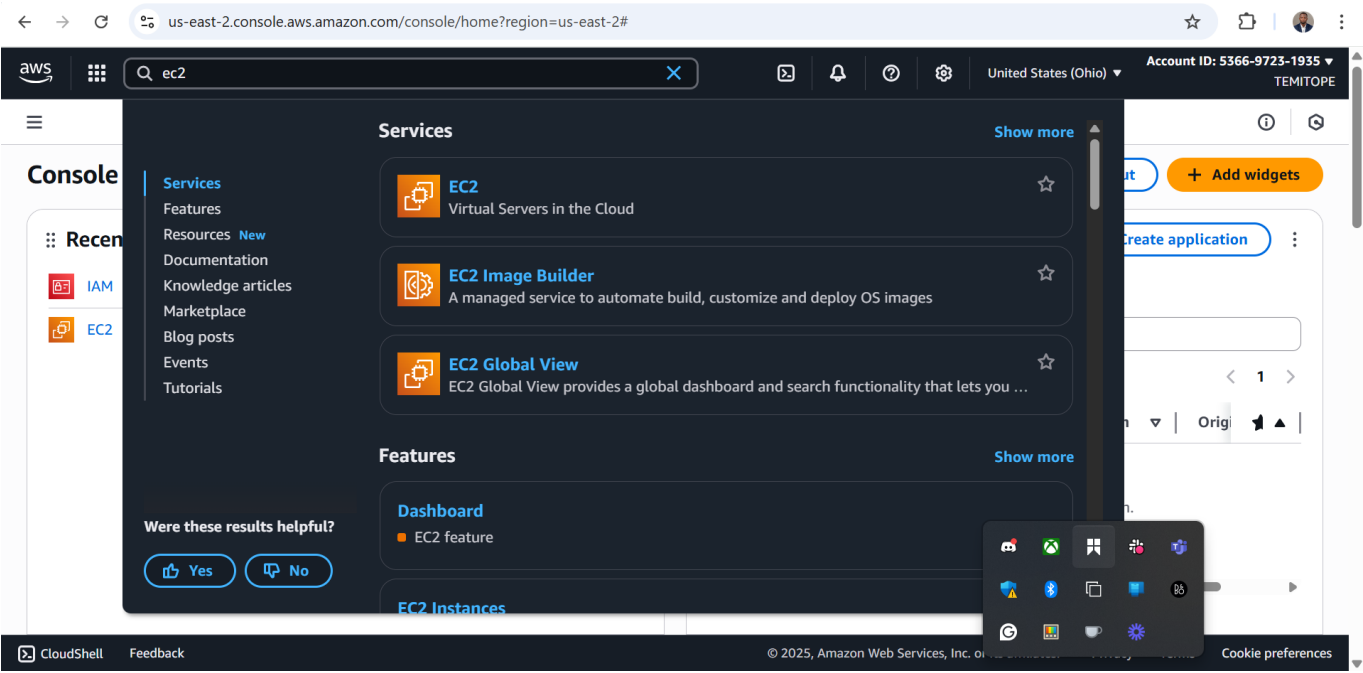
Step 17 — Log in to your AWS Management Console

If you don't have an account, you will have to create one. By gaining access to the AWS Console, we can create resources.



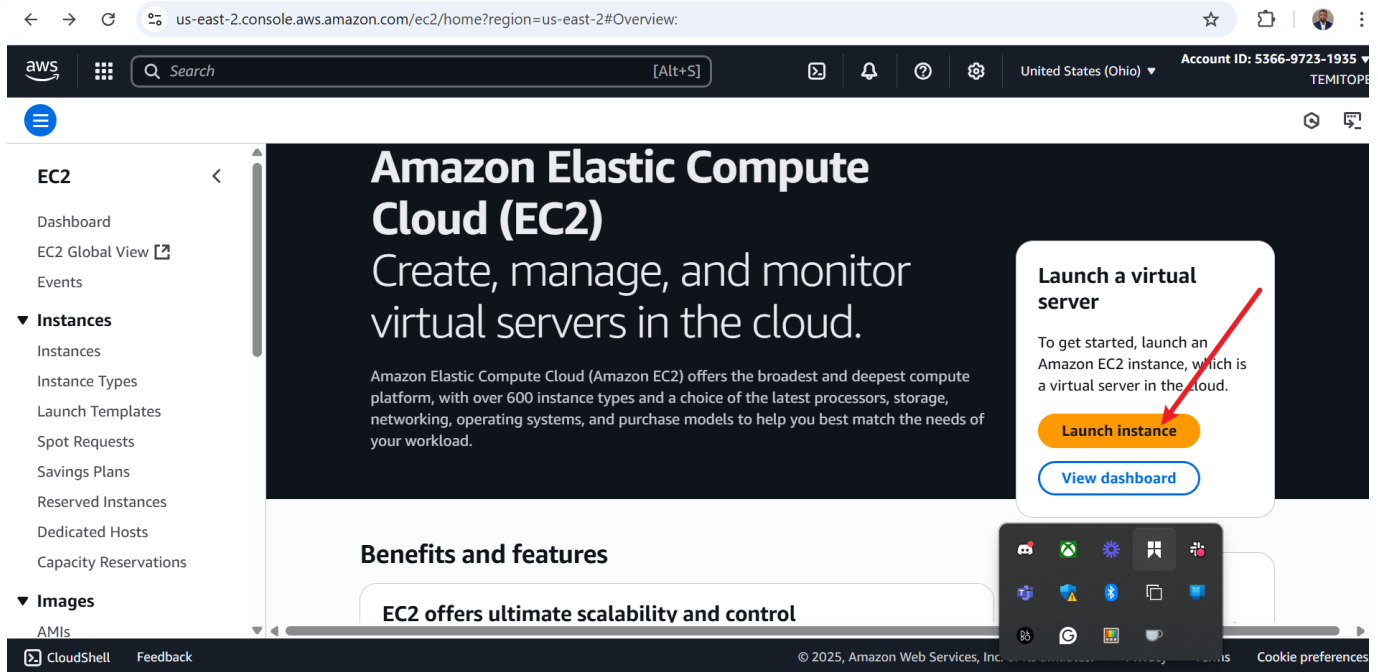
Step 18 — Open EC2 service

Use the console search to locate EC2 and open the EC2 Dashboard.



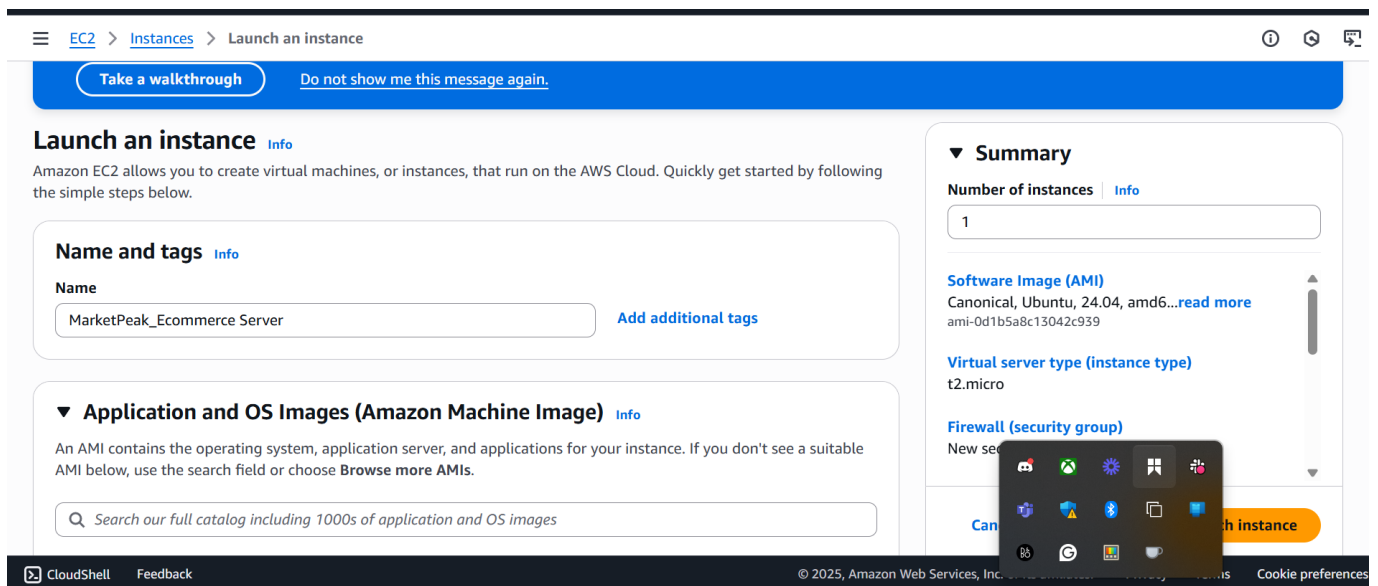
Step 19 — Launch a new EC2 instance

Click "Launch instance" to start the instance wizard.



Step 20 — Name your instance `MarketPeak_Ecommerce_Server`

Use a clear resource tag for easy identification.



Step 21 — Choose Ubuntu Server 24.04 LTS AMI

A current, stable Ubuntu LTS provides security patches and compatibility with Ubuntu packages.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LaunchInstances:

aws Search [Alt+S] United States (Ohio) Account ID: 5366-9723-1935 TEMITOPE

EC2 > Instances > Launch an instance

Amazon Linux macOS **Ubuntu** Windows Red Hat SUSE Linux

Browse more AMIs Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type Free tier eligible

ami-0d1b5a8c13042c939 (64-bit (x86)) / ami-019eef96c2865995 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture AMI ID Publish Date Username

64-bit (...) ami-0d1b5a8c13042c939 2025-06-10 ubuntu Verified provider

Summary

Number of instances Info 1

Software Image (AMI) Canonical, Ubuntu, 24.04, amd64...read more ami-0d1b5a8c13042c939

Virtual server type (instance type) t2.micro

Firewall (security group) New security group

Launch instance Review code

CloudShell Feedback © 2025, Amazon Web Services, Inc. Cookie preferences

Step 22 — Choose instance type (t2.micro for free-tier)

t2.micro is free-tier eligible and sufficient for demo deployments.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LaunchInstances:

aws Search [Alt+S] United States (Ohio) Account ID: 5366-9723-1935 TEMITOPE

EC2 > Instances > Launch an instance

Instance type Info | Get advice

Instance type

t2.micro Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

Additional costs apply for AMIs with pre-installed software

All generations Compare instance types

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Launch instance Review code

CloudShell Feedback © 2025, Amazon Web Services, Inc. Cookie preferences

Step 23 — Create and download an SSH key pair

Create a new key pair, download the `.pem` private key and keep it secure. Save it in a path you will use to SSH (e.g., `~/ .ssh/marketpeak.pem`).

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LaunchInstances:

aws Search [Alt+S] United States (Ohio) Account ID: 5366-9723-1935 TEMITOPE

EC2 > Instances > Launch an instance

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

MarketPeak_Ecommerce_key

Create new key pair

Summary

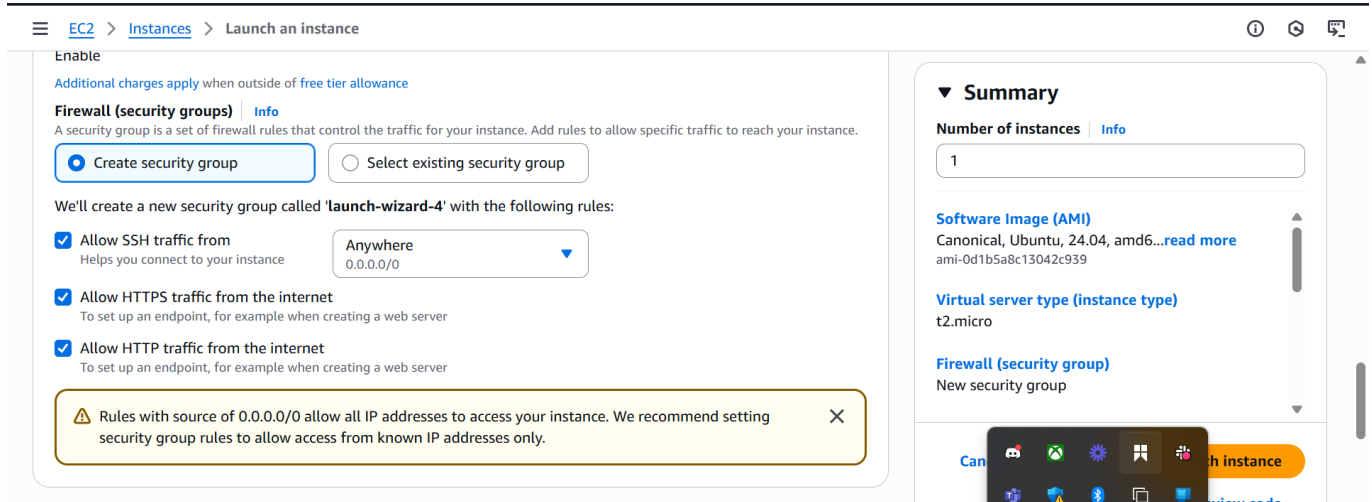
Number of instances Info 1

Software Image (AMI) Canonical, Ubuntu, 24.04, amd64...read more ami-0d1b5a8c13042c939

Step 24 — Configure security group: allow SSH, HTTP, HTTPS

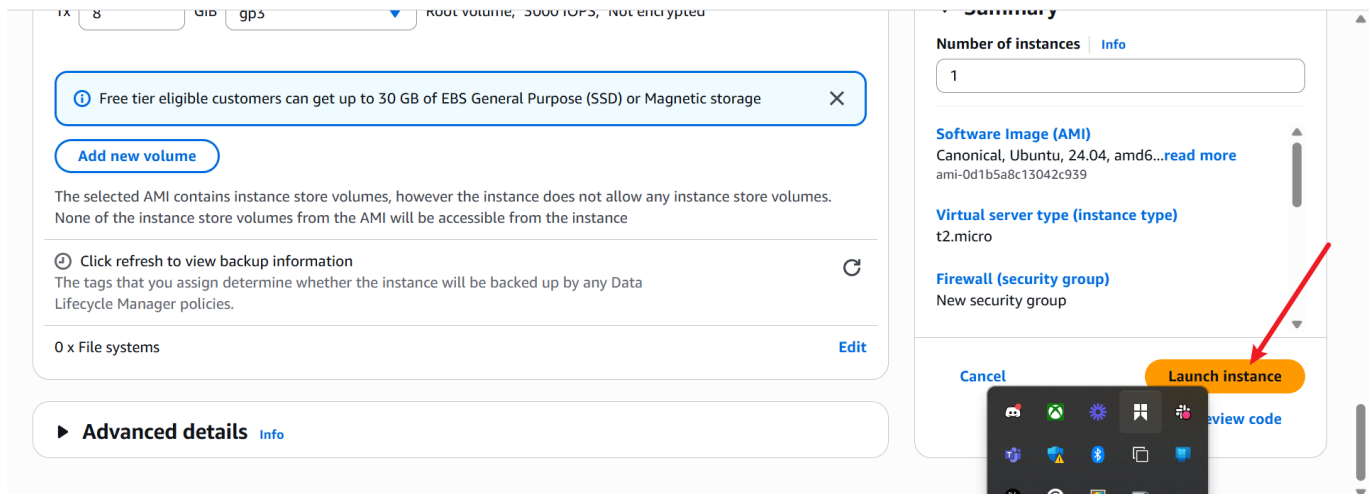
Add inbound rules for (allow for all):

- SSH (TCP 22) — Source: your IP (or 0.0.0.0/0 for testing, not recommended)
- HTTP (TCP 80) — Source: 0.0.0.0/0
- HTTPS (TCP 443) — Source: 0.0.0.0/0



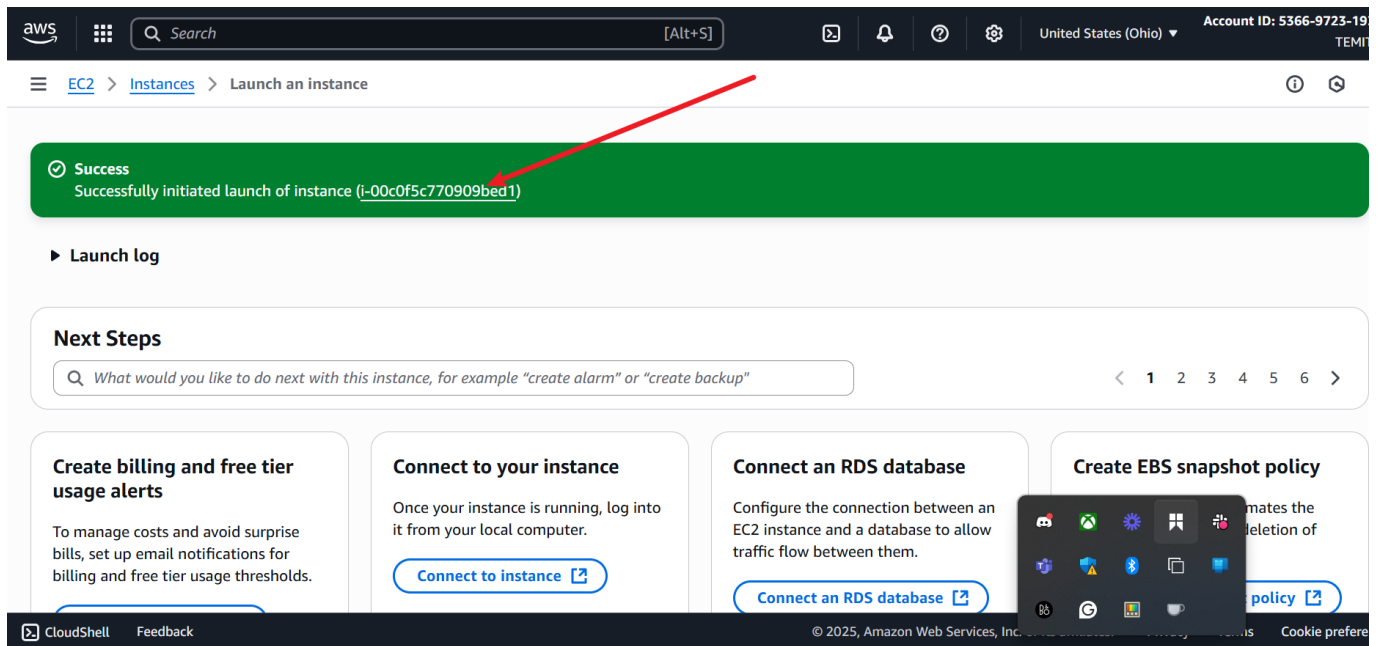
Step 25 — Launch the EC2 instance

Review settings and click Launch. Instance enters **pending** then **running** state.



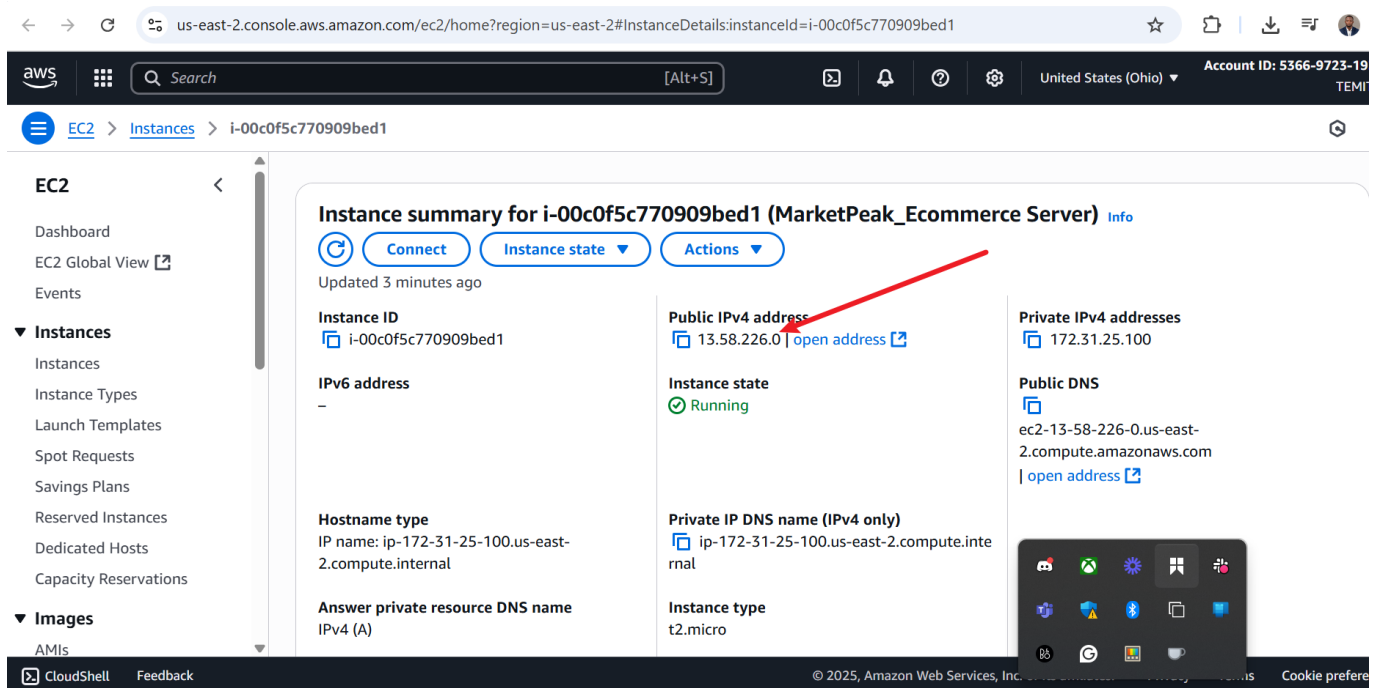
Step 26 — Click the instance ID in the success message

Use the link in the launch-success banner to open the instance details page.



Step 27 — Note the public IPv4 address & click Connect

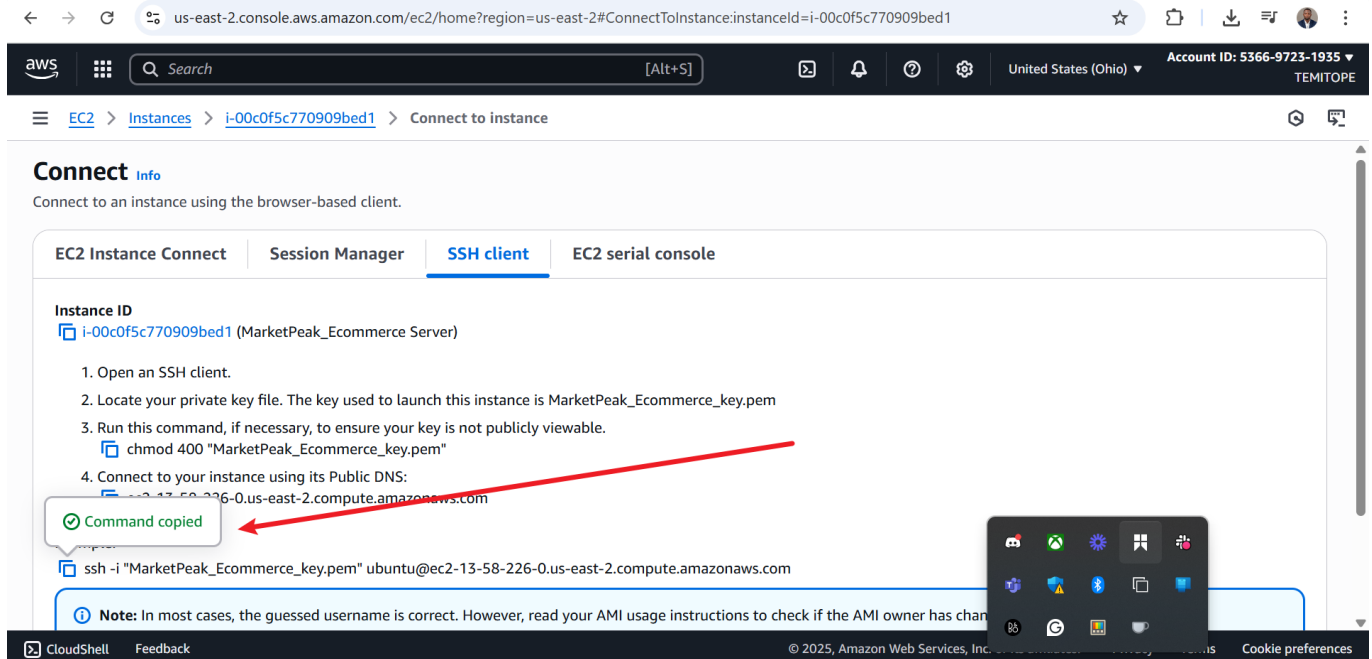
Copy the public IP (or public DNS) for SSH access.



Step 28 — Copy the example SSH command from the Connect page

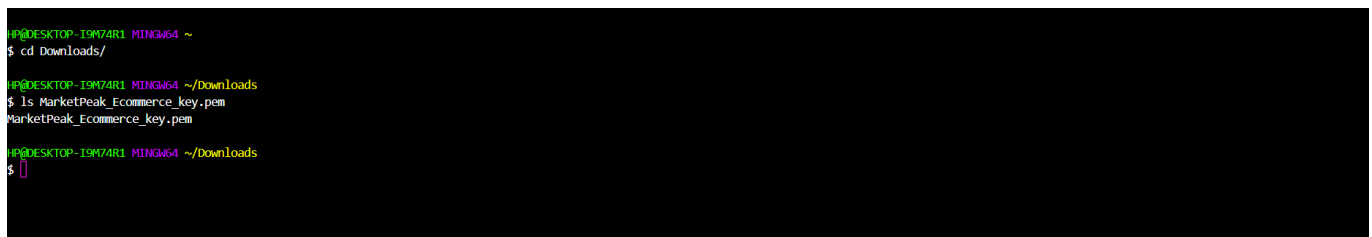
- Example provided by AWS:

```
ssh -i "your-key-pair.pem" ubuntu@<PUBLIC_IP>
```



Step 29 — Locate downloaded .pem on your machine

Confirm the path where the key was saved (Downloads, ~/.ssh, etc.).

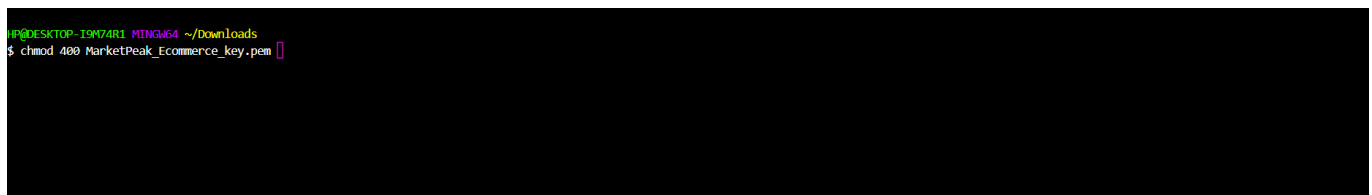


Step 30 — Set proper permissions on the key

- **Command:**

```
chmod 400 /path/to/your-key-pair.pem
```

SSH requires private key permissions to be restricted.



Step 31 — Connect to the instance via SSH

- **Command (example):**

```
ssh -i "~/Downloads/your-key-pair.pem" ubuntu@<PUBLIC_IP>
```

```
HP@DESKTOP-19M74R1 MINGW64 ~/Downloads  
$ ssh -i "MarketPeak_Ecommerce_key.pem" ubuntu@ec2-13-58-226-0.us-east-2.compute.amazonaws.com
```

Step 32 — Accept SSH fingerprint (type **yes**)

First-time connection will show the host fingerprint; type **yes** to store it.

```
MINGW64/c/Users/HP/Downloads  
  
HP@DESKTOP-19M74R1 MINGW64 ~/Downloads  
$ ssh -i "MarketPeak_Ecommerce_key.pem" ubuntu@ec2-13-58-226-0.us-east-2.compute.amazonaws.com  
The authenticity of host 'ec2-13-58-226-0.us-east-2.compute.amazonaws.com (13.58.226.0)' can't be established.  
ED25519 key fingerprint is SHA256:dWwv+9HcwhvMRnJJZ7sn0UrRyyvy3pgQyS8XaY9xfmw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

Step 33 — Confirm you are connected to the EC2 instance

You'll see something like `ubuntu@ip-...:~$`

```
ED25519 key fingerprint is SHA256:dtWv+9HcwhvMnJZ7sn0UrYyy3pgQyS8XaV9xfmw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-58-226-0.us-east-2.compute.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1029-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Aug 10 21:00:37 UTC 2025

System load:  0.08          Processes:      103
Usage of /:   25.4% of 6.71GB Users logged in:   0
Memory usage: 19%          IPv4 address for enx0: 172.31.25.100
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-25-100:~$
```

You can also confirm by running `uname -a` or `lsb_release -a` to confirm Ubuntu OS.

Step 34 — On GitHub, open repo and click "Code" to get clone options

Navigate to your repo and reveal HTTPS/SSH options.

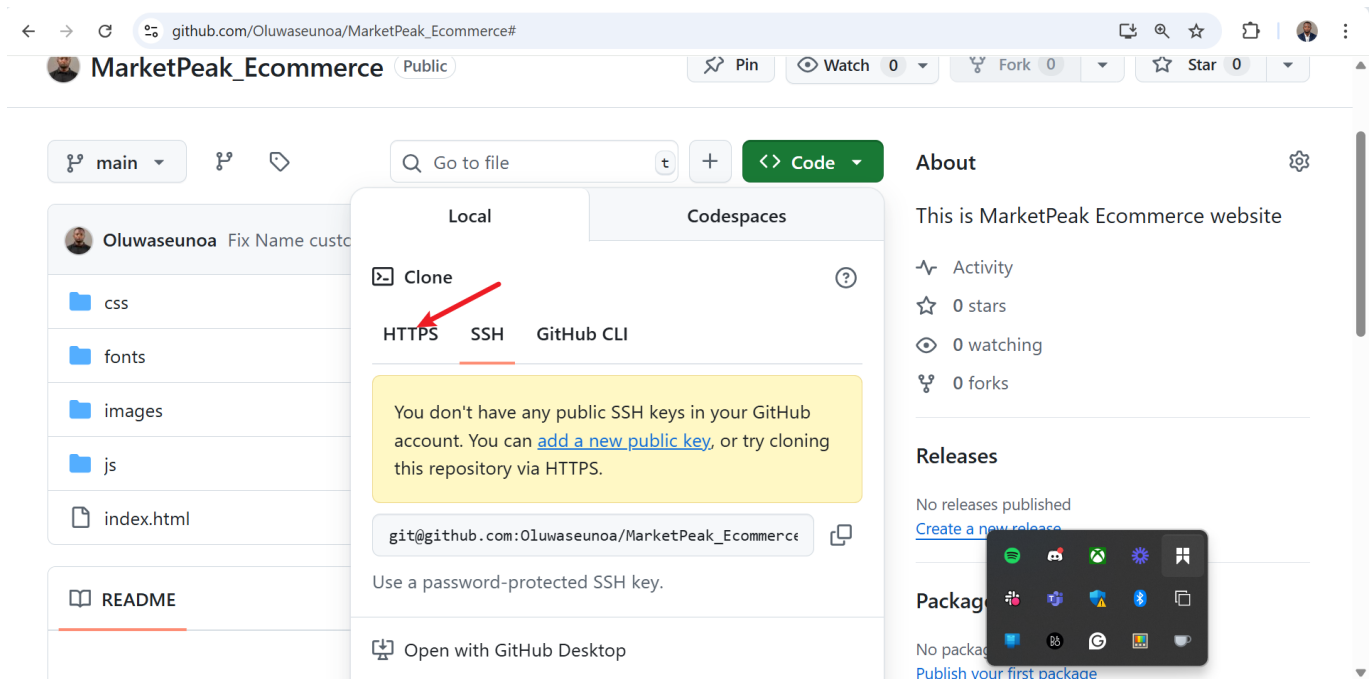
The screenshot shows the GitHub interface for the repository 'MarketPeak_Ecommerce' by user 'Oluwaseunola'. The 'Code' button is highlighted with a red arrow. The repository is public and has 0 watches, 0 forks, and 0 stars. The commit history table shows the following commits:

File	Commit Message	Time
css	Initial commit of with basic ecom...	yesterday
fonts	Initial commit of with basic ecom...	yesterday
images	Initial commit of with basic ecom...	yesterday
js	Initial commit of with basic ecom...	yesterday

The 'About' section on the right states: 'This is MarketPeak Ecommerce website'. Below this, it shows 'Activity' with 0 stars, 0 watching, and 0 forks. The 'Releases' section shows 'No releases published'.

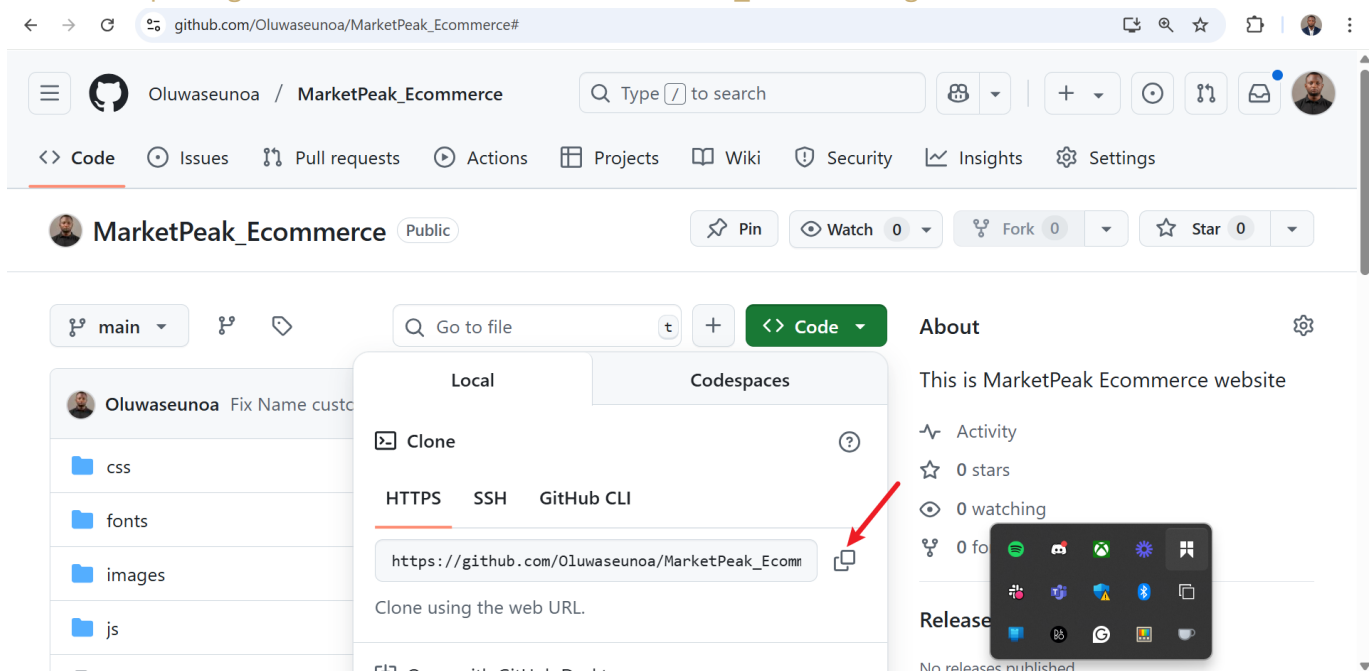
Step 35 — Choose HTTPS (or SSH if configured)

Going for HTTPS because it requires no additional server-side key configuration; SSH requires adding public key to GitHub account.



Step 36 — Copy the HTTPS clone URL

Mine is https://github.com/Oluwaseunola/MarketPeak_Ecommerce.git



Step 37 — Clone the repository on the EC2 instance and confirm the download

- **Commands:**

```
git clone https://github.com/Oluwaseunola/MarketPeak_Ecommerce.git
ls
```

The repo appears as a folder in your home directory.

```
ubuntu@ip-172-31-25-100:~$ git clone https://github.com/Oluwaseunola/MarketPeak_Ecommerce.git
Cloning into 'MarketPeak_Ecommerce'...
remote: Enumerating objects: 42, done.
remote: Counting objects: 100% (42/42), done.
remote: Compressing objects: 100% (40/40), done.
remote: Total 42 (delta 2), reused 42 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (42/42), 2.77 MiB | 2.08 MiB/s, done.
Resolving deltas: 100% (2/2), done.
ubuntu@ip-172-31-25-100:~$ ls MarketPeak_Ecommerce/
css  fonts  images  index.html  js
ubuntu@ip-172-31-25-100:~$
```

Step 38 — Update and upgrade Ubuntu packages

- **Commands:**

```
sudo apt update && sudo apt upgrade -y
```

This keep packages up-to-date for security and compatibility.

```
ubuntu@ip-172-31-25-100:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1315 kB]
Get:6 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [264 kB]
Get:7 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [163 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [377 kB]
]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1650 k
B]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [361 k
B]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212
B]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940
B]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7060 B]
Get:14 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [30.
7 kB]
Get:15 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [2
16 B]
Get:16 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [2
12 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 4622 kB in 2s (2968 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
105 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
```

Step 39 — Install Apache2 web server

- **Command:**

```
sudo apt install apache2 -y
```

```
MINGW64:/c/Users/HP/Documents
ubuntu@ip-172-31-25-100:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.7).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-25-100:~$
```

Step 40 — Start and enable Apache2

- **Commands:**


```
sudo systemctl start apache2
sudo systemctl enable apache2
```

```
ubuntu@ip-172-31-25-100:~$ sudo systemctl start apache2
ubuntu@ip-172-31-25-100:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-25-100:~$
```

Step 41 — Prepare web directory: move or link repo to `/var/www/html`

- **Options:**
 - **Copy (recommended for static sites):**

```
sudo rm -rf /var/www/html/*
sudo cp -r ~/MarketPeak_Ecommerce/* /var/www/html/
```

- **Alternative — symlink (easier for development):**

```
sudo rm -rf /var/www/html
sudo mv ~/MarketPeak_Ecommerce /var/www/
sudo ln -s /var/www/MarketPeak_Ecommerce /var/www/html
sudo chown -R www-data:www-data /var/www/MarketPeak_Ecommerce
```

```
MINGW64:/c/Users/HP/Documents

ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$ sudo rm -rf /var/www/html
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$ sudo mv ~/MarketPeak_Ecommerce /var/www/
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$ sudo ln -s /var/www/MarketPeak_Ecommerce /var/www/html
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$ sudo chown -R www-data:www-data /var/www/MarketPeak_Ecommerce
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$
ubuntu@ip-172-31-25-100:~/MarketPeak_Ecommerce$
```

Step 42 — Reload Apache to pick up changes

Reload picks up file changes without restarting the server.

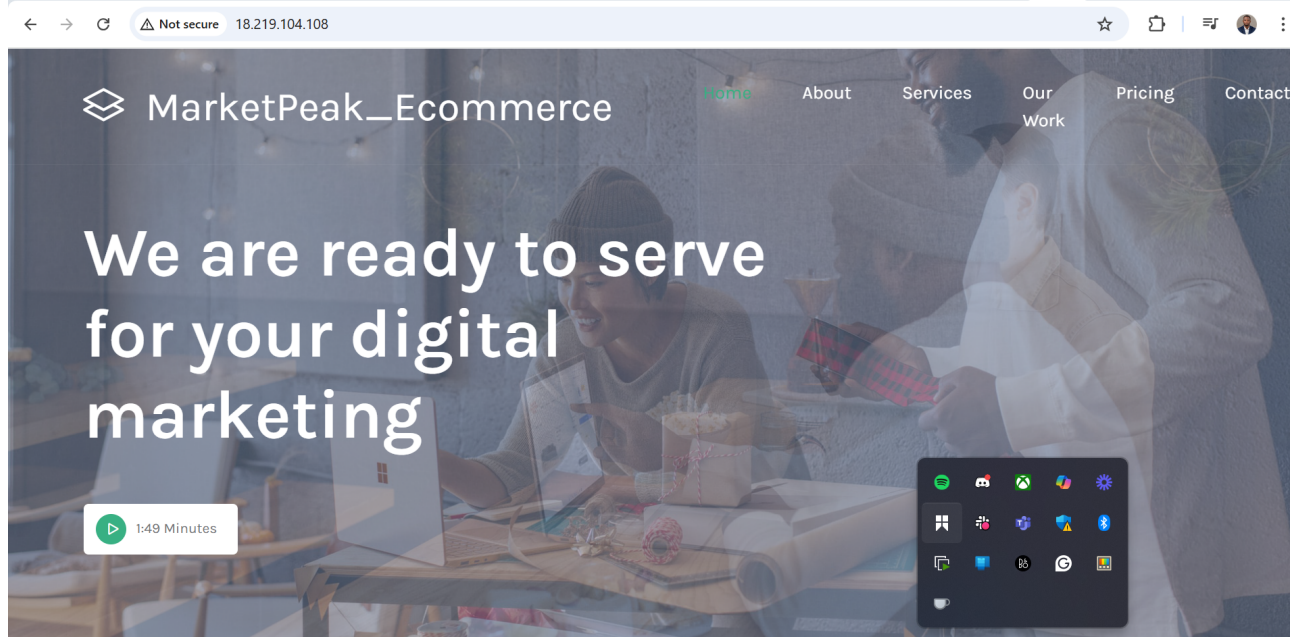
- **Command:**

```
sudo systemctl reload apache2
```

```
ubuntu@ip-172-31-25-100:~$ sudo systemctl reload apache2
ubuntu@ip-172-31-25-100:~$
```

Step 43 — Open the site in a browser using the public IP

- **URL:** http://<PUBLIC_IP>/ (or the server domain if you configured DNS) The MarketPeak site should load and show the customized branding.



(Steps 44–58 — development workflow, PRs, merge & deploy)

The document continues with the development branch workflow, making a change on a **development** branch, pushing, opening a pull request, merging into **main**, pulling changes on the server, and verifying updates live on the site. Steps 44–58 include creating a **development** branch, changing the site title to **MarketPeak_4.0**, staging, committing, pushing, creating PR, merging, pulling on server, reloading Apache, and confirming the change in the browser. For completeness these steps are fully documented below (with commands and checks):

Step 44 — Create development branch for changes

This isolate feature work from **main**.

- **Commands:**

```
git checkout -b development
```

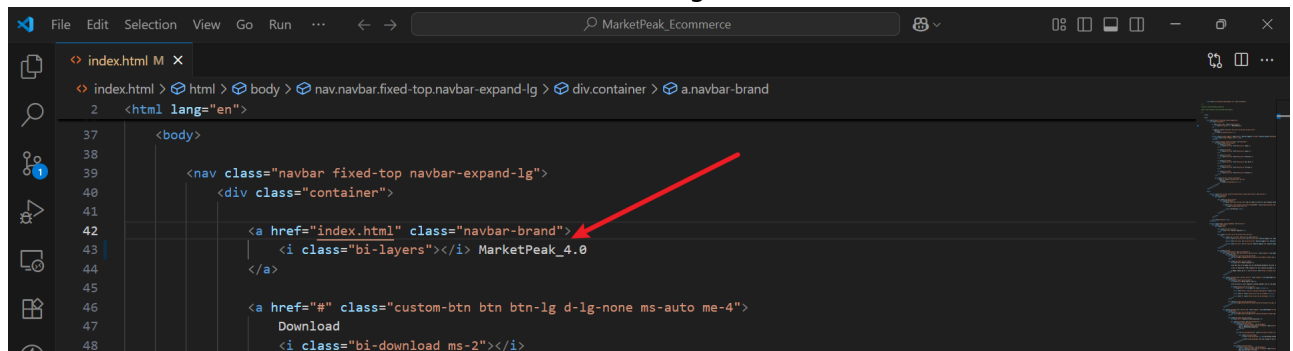


```
$ git branch development
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (main)
$ git checkout development
Switched to branch 'development'
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (development)
$
```

Step 45 — Make a change (example: update site title)

Edit `index.html`. I am changing the website `<title>` from `MarketPeak_Ecommerce` to `MarketPeak_4.0`.

- **Command (editor):** code `index.html` then save changes.



```
index.html M X
index.html > html > body > nav.navbar.fixed-top.navbar-expand-lg > div.container > a.navbar-brand
2 <html lang="en">
37 <body>
38
39 <nav class="navbar fixed-top navbar-expand-lg">
40 <div class="container">
41
42 <a href="index.html" class="navbar-brand">
43 <i class="bi-layers"></i> MarketPeak_4.0
44 </a>
45
46 <a href="#" class="custom-btn btn btn-lg d-lg-none ms-auto me-4">
47 Download
48 <i class="bi-download ms-2"></i>
```

Step 46 — Stage the change

- **Command:**

```
git add .
```

- **Verification:** `git status` shows staged files.

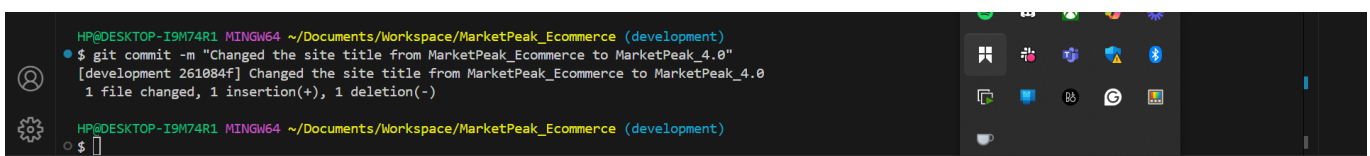


```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (development)
$ git add .
```

Step 47 — Commit change with a clear message

- **Command:**

```
git commit -m "Changed the site title from MarketPeak_Ecommerce to
MarketPeak_4.0"
```

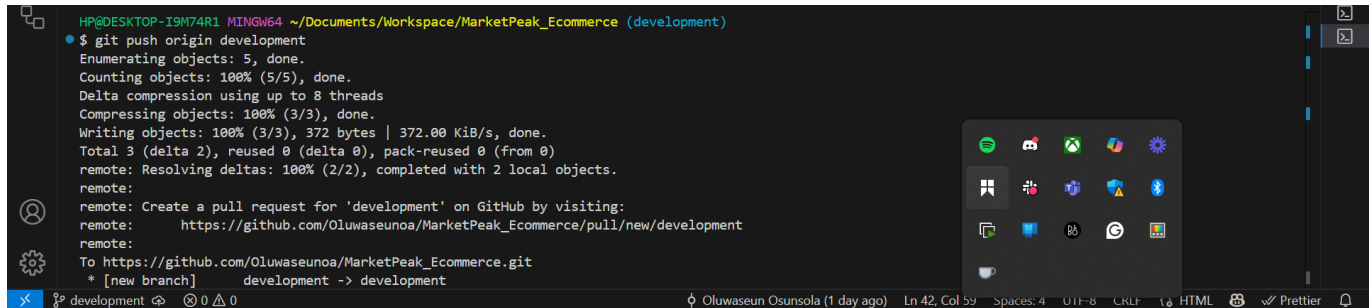


```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (development)
$ git commit -m "Changed the site title from MarketPeak_Ecommerce to MarketPeak_4.0"
[development 261084f] Changed the site title from MarketPeak_Ecommerce to MarketPeak_4.0
1 file changed, 1 insertion(+), 1 deletion(-)
```

Step 48 — Push development branch to origin

- **Command:**

```
git push origin development
```

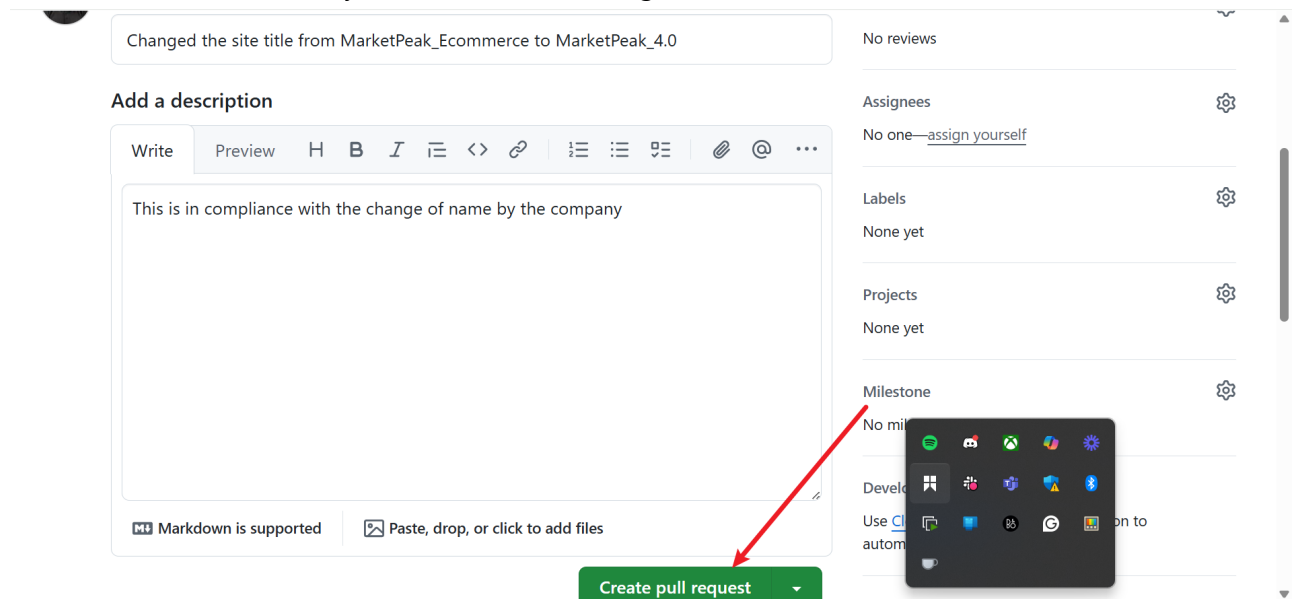


```
HP@DESKTOP-I9M74R1 MINGW64 ~/Documents/Workspace/MarketPeak_Ecommerce (development)
$ git push origin development
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 372 bytes | 372.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'development' on GitHub by visiting:
remote:   https://github.com/OluwaseunOa/MarketPeak_Ecommerce/pull/new/development
remote:
remote: To https://github.com/OluwaseunOa/MarketPeak_Ecommerce.git
 * [new branch]      development -> development
```

Step 49 — Create a Pull Request (Development → Main) on GitHub

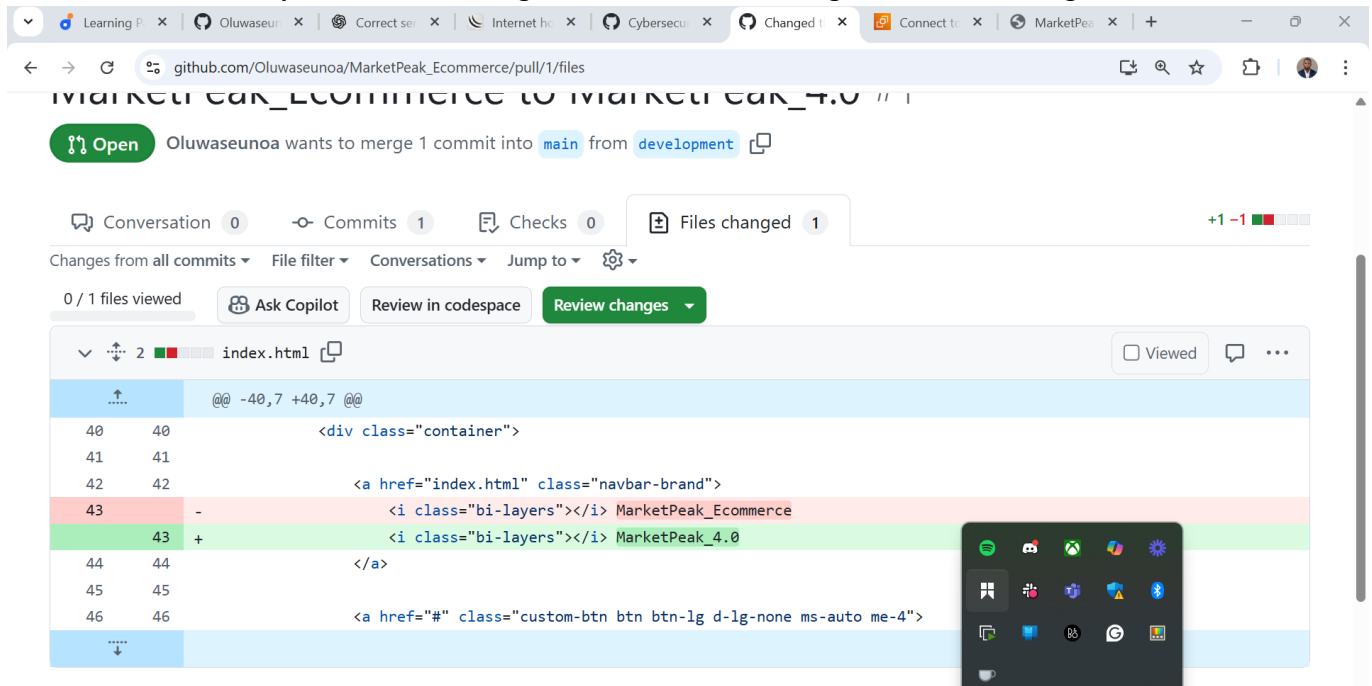
In GitHub UI click "Compare & pull request" and open a PR with a descriptive title and summary.

- **Checklists:** Ensure diffs only include intended changes.



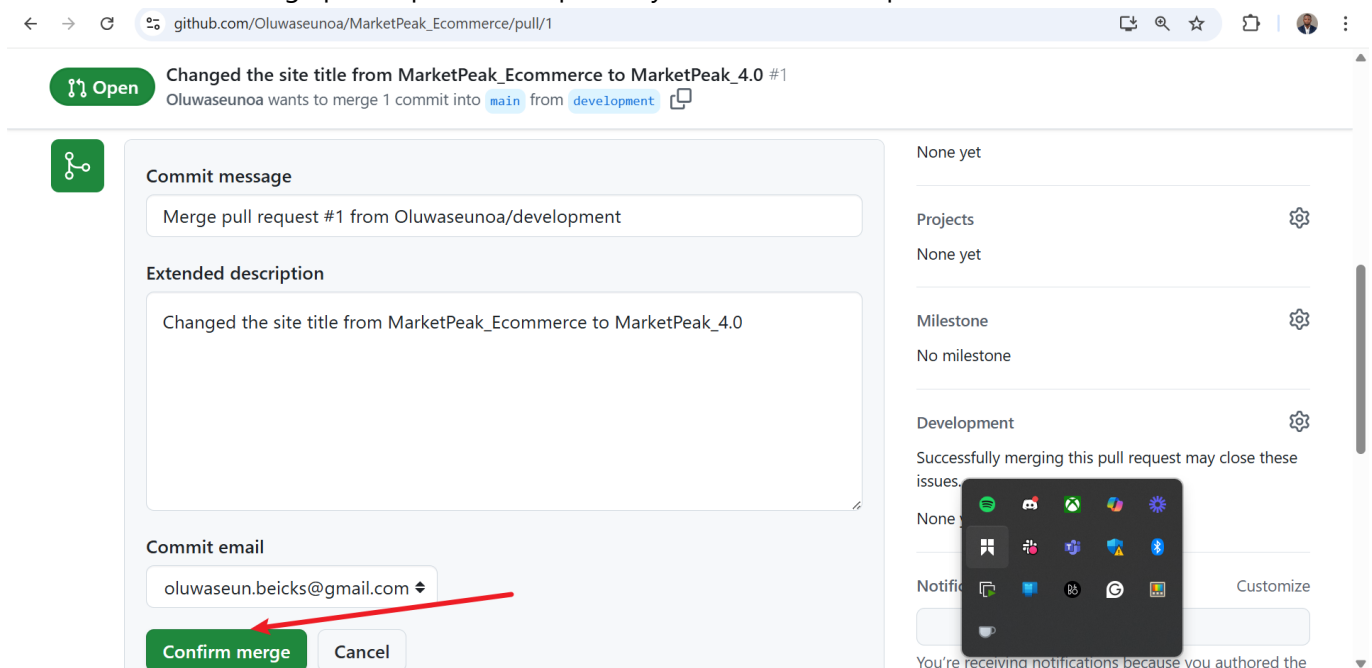
Step 50 — Review changes and run checks before merging

Review the diff, run any CI checks (if configured), and confirm changes are safe to merge.



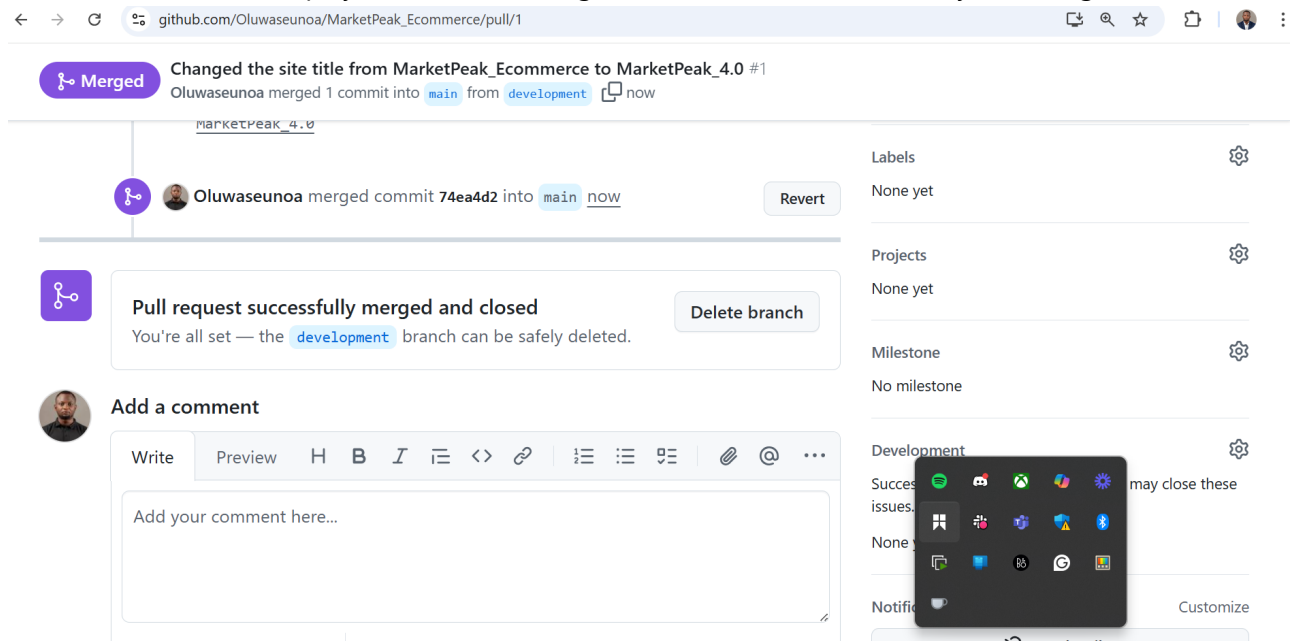
Step 51 — Merge the Pull Request to main

Use GitHub UI to "Merge pull request" and optionally delete the development branch.



Step 52 — Confirm successful merge

- **Verification:** GitHub displays a successful merge banner; **main** now contains your changes.

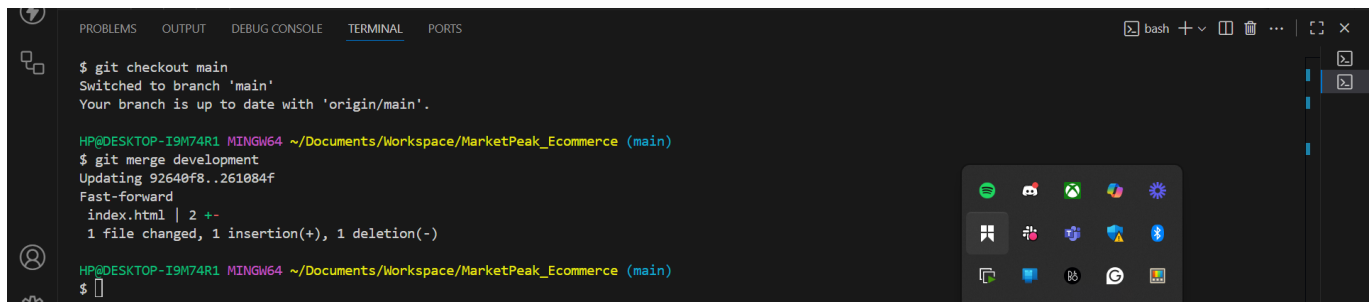


Step 53 — Pull the updated **main** on any environment that serves production

This will keep production files in sync with repo.

- **Commands (on server or local):**

```
git checkout main
git merge development
```

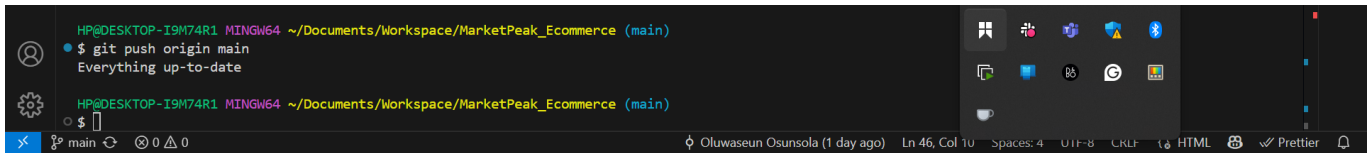


Step 54 — Push main (if you merged locally)

- **Command (if merging locally):**

```
git push origin main
```

This will say 'Everything up-to-date'



Step 55 — Update production server files (if using clone in `/var/www`)

- **If you copied files to `/var/www/html` earlier:** re-copy or pull changes into that directory. Example using a git-backed setup:

```
cd /var/www/html
sudo git pull origin main
```

- **Alternative (if using symlink):** Changes in the home clone are visible immediately.
- **Check:** `cat /var/www/html/index.html | grep "MarketPeak_4.0"` to confirm the updated title exists.

Step 56 — Fix permissions & ownership for Apache

- **Commands:**

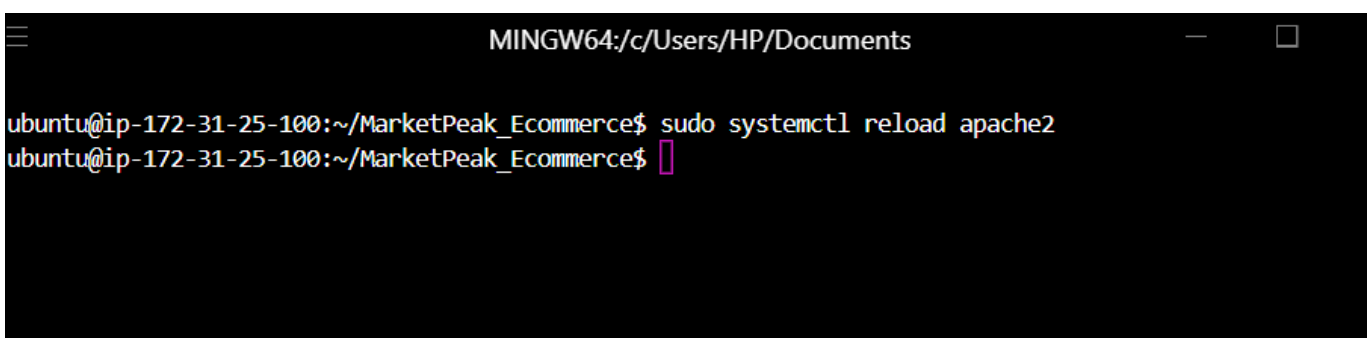
```
sudo chown -R www-data:www-data /var/www/html
sudo find /var/www/html -type d -exec chmod 755 {} \;
sudo find /var/www/html -type f -exec chmod 644 {} \;
```

Ensures Apache can read files and provides secure permissions.

Step 57 — Reload Apache to apply updates

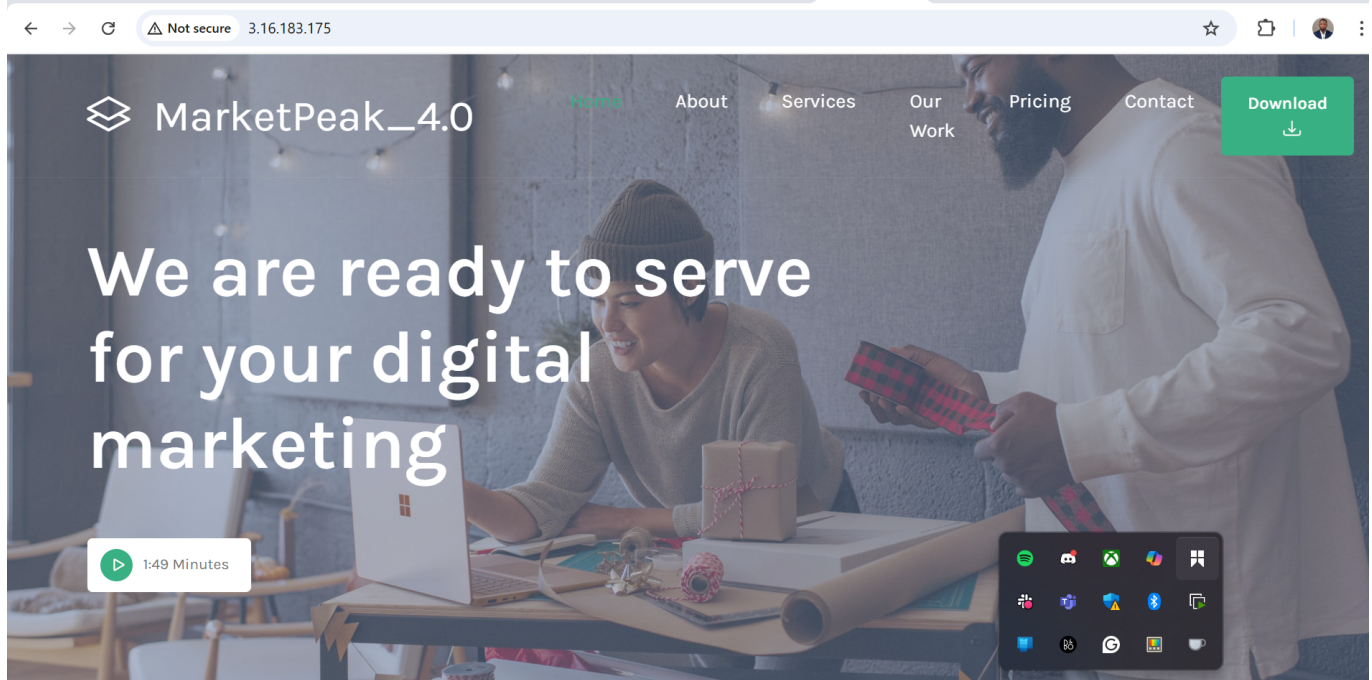
- **Command:**

```
sudo systemctl reload apache2
```



Step 58 — Verify changes on live website

Action: Open `http://<PUBLIC_IP>/` in a browser and confirm the `MarketPeak_4.0` title (or other changes) appears.



Troubleshooting (common issues & fixes)

- **fatal: detected dubious ownership in repository** — run `git config --global --add safe.directory /path/to/repo` on the server if you trust the repo owner. See screenshot examples in other tasks.
- **Permission denied (publickey)** — ensure you used the correct user (`ubuntu` for Ubuntu AMIs), correct `.pem` and `chmod 400` was applied.
- **403 Forbidden on Apache** — missing `index.html` in `/var/www/html`, wrong permissions, or Apache `Require all denied` in config. Check `/var/log/apache2/error.log`.

Deliverables & Submission

- `README.md` (this document) committed to GitHub.
- `img/` folder containing the 58 screenshots.
- Live site URL: `http://<EC2_PUBLIC_IP>/` (replace with your IP or domain).

(Steps 8–43 would continue in similar detail, with each step explaining the purpose, commands/actions, expected result, and linking to the screenshot.)

Step 44: Add Product Images

Action: Replace placeholder images in the `img/` folder with your actual product images.

- **Detail:** Ensure filenames match those referenced in the HTML/CSS.
- **Verification:** Refresh the website to confirm new images display correctly.

Step 45: Update Product Descriptions

Action: Edit relevant HTML sections to insert accurate product descriptions.

Step 46: Configure Navigation Menu

Action: Update links in the header to point to the correct sections/pages.

Step 47: Add Contact Information

Action: Edit the contact section with actual email, phone, and address.

Step 48: Test Contact Form

Action: Configure the backend script or service (e.g., Formspree) to handle form submissions, then submit a test message.

Step 49: Configure Footer

Action: Update footer text, links, and copyright info.

Step 50: Optimize for Mobile

Action: Test on mobile devices/emulators, adjust CSS media queries if needed.

Step 51: Test Across Browsers

Action: Open the site in Chrome, Firefox, Edge, and Safari to ensure consistent appearance.

Step 52: Check Performance with Lighthouse

Action: Run Google Lighthouse audit and note performance, accessibility, and SEO scores.

Step 53: Minify CSS and JS

Action: Use tools like [uglify-js](#) and [cssnano](#) to reduce file sizes.

Step 54: Push Project to GitHub

Action:

```
git add .
git commit -m "Initial project setup and customization"
git branch -M main
git remote add origin <your-repo-url>
git push -u origin main
```

Step 55: Deploy to Hosting Service

Action: Deploy to GitHub Pages, Netlify, or Vercel.

Step 56: Test Live Deployment

Action: Visit the live URL and test all links and forms.

Step 57: Final Review and QA

Action: Conduct a final review with the team/client, note issues, and fix.

Step 58: Project Completion Confirmation

Action: Document the completion and store all assets and source code in a safe backup.